

RASTA AI

SIH 2026 — Day 1 Task 1

Problem Understanding, Solution Planning & System Design

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How to read this document. Every capability carries one of six status labels: **IMPLEMENTED**, **PARTIALLY IMPLEMENTED**, **NOT IMPLEMENTED**, **BROKEN**, **BLOCKED**, **NEEDS TESTING**. A label is given only when the evidence named beside it exists in the repository or in a certification log. Where evidence is missing the entry says **NOT VERIFIED**. No accuracy percentage is claimed anywhere in this document, because no predictive model has been evaluated on a held-out set (`docs/AI_MODELS.md` §0).

Table of contents

Table of contents	2
1. Problem Statement Understanding	4
1.1 Problem Context	4
1.2 Existing System Limitations	4
1.3 Core Problem	5
1.4 Stakeholder Pain Points	5
1.5 Why Existing Approaches Are Insufficient	5
1.6 Design Constraints	5
2. Proposed Solution	7
2.1 Solution Overview	7
2.2 Manager Platform	7
2.3 Driver Application	8
2.4 Backend Platform	8
2.5 Intelligent / AI Layer	9
2.6 Offline / Connectivity-Aware Strategy	10
3. Unique Key Features	11
3.1 Validated Route Planning (real roads, checked answers)	11
3.2 Planned Route vs Observed GPS Track	11
3.3 Real Route Alternatives (no fabricated backups)	11
3.4 Evidence-Based Route Risk (UNKNOWN ≠ SAFE)	12
3.5 Route Governance: Review Authorisation and the Dispatch Gate	12
3.6 Pickup-First Navigation with Turn Guidance	13
3.7 Off-Route Detection and Governed Reroute	13
3.8 Connectivity-Aware Driver App and GPS Queue	13
3.9 Pre-download / Offline Trip Package	14
3.10 Weather, Landslide and Disruption Awareness on the Phone	14
3.11 Driver Safety Assistant (offline first, online wording optional)	14
3.12 Emergency Assistance and Fleet Sentinel	14
3.13 Roadside Facilities	15
3.14 Trip Lifecycle with Post-Pickup Cargo Rules	15
3.15 Fleet Monitoring and Manager–Driver Coordination	15
3.16 Battery / Last-Known-Location Strategy	15
3.17 Multilingual Assistance	15
3.18 One-to-One Driver–Truck Assignment with Physical Verification	15
4. Target Users	16
4.1 Fleet Manager (dispatcher)	16

4.2 Truck Driver	16
4.3 Logistics Company / Fleet Operator	16
4.4 Emergency / Support Personnel	16
5. System Architecture	18
5.1 High-Level Architecture	18
5.2 Client Layer	18
5.3 Backend Layer	19
5.4 Data Layer	19
5.5 External Services	21
5.6 Architecture Principles	22
6. Workflow / User Flow Diagram	23
6.1 Manager Workflow (as the hosted console behaves, certified 16 Sep 2026)	23
6.2 Driver Workflow (physical phone, APK 1.0.18, certified 16 Sep 2026)	23
6.3 Emergency Flow	24
7. Data Flow Diagram	25
DFD Level 0 (context)	25
DFD Level 1 (processes)	26
8. Module Definition	28
9. Technology Stack	30
Frontend — Manager	30
Driver Application	30
Backend	30
Database	30
Mapping / Routing	31
Weather / Risk Sources	31
AI providers	31
DevOps / Tooling	31
10. Technology Justification	32
11. Scalability Analysis	33
12. Security	34
13. Failure & Degradation Strategy	35
14. Implementation Status Matrix	37
15. Final System Flowchart	39
16. Completion Report	40

1. Problem Statement Understanding

1.1 Problem Context

Freight movement in the North Eastern Region (NER) of India runs over a small number of long, hilly corridors: Guwahati–Shillong, Guwahati–Silchar, Guwahati–Itanagar, Dimapur–Kohima–Imphal. These roads share operating conditions that mainstream logistics tooling was not designed for.

- **Difficult road conditions.** Steep gradients, single-carriageway hill sections and monsoon damage. The project's own terrain sampling (Copernicus DEM through Open-Meteo, SRTM through OpenTopoData) is used precisely because gradient matters on these corridors (backend/app/services/terrain.py).
- **Long-distance transportation with few alternatives.** Measured against the live routing provider on eight real NER corridors, six have exactly one sensible road and two have a second corridor; never three (comment in manager-web/src/components/TripRouteReview.tsx). When the one road is cut, the truck waits.
- **Route disruption from landslides and floods.** The NASA Global Landslide Catalog records repeated slides within 5 km of these roads (the project ships a 2007–2017 snapshot; backend/app/services/landslide/history.py). River discharge on the Brahmaputra tributaries drives seasonal closures (GloFAS discharge context, backend/app/domain/flood.py).
- **Extreme weather.** Monsoon rainfall intensities above 3 mm/h on a corridor point are the project's own rain-exposure trigger (backend/app/domain/route_risk.py, MAX_RAIN_POINTS = 45).
- **Weak or intermittent connectivity.** A truck loses mobile data for long stretches; signal bars do not mean data. The driver application therefore derives its connection status from whether its last server poll succeeded, never from the operating system's "online" flag (driver-app/src/trip/TripProvider.tsx, isStale).
- **Lack of timely driver information.** Official disaster alerts (NDMA SACHET) are published as a national CAP feed; nothing filters them to the district a specific truck is entering.
- **Emergency situations.** A stationary truck on a hill road for an hour with no contact is either a rest stop or an accident; a fleet office cannot tell which without a check-in mechanism.
- **Route visibility for fleet managers.** Managers see a planned corridor but not the observed track, so a truck that has left the road is invisible until it phones in.
- **Driver safety and language.** Drivers on NER corridors speak Assamese, Bengali, Bodo, Manipuri, Nepali, Hindi and English; safety guidance in English only is guidance for some.
- **Inefficient routing and dependency on continuous internet.** Consumer navigation assumes the internet is always there and that the shortest road is the best road.

1.2 Existing System Limitations

Conventional navigation and fleet tools are excellent at what they were built for. The gaps below are specific to this use case, not criticism of those products.

Limitation	What it means on an NER corridor
Internet-dependent information	Map tiles, rerouting and alerts stop the moment data drops. The driver is left with a frozen screen exactly where the road is hardest.
Limited route-risk awareness	Consumer routing optimises time and distance. It has no notion of landslide history, river discharge, terrain gradient or an active district alert.
Disconnected logistics tools	Trip planning, driver assignment, vehicle tracking and safety information usually live in separate systems (or a WhatsApp group).
Weak offline preparedness	Nothing is pre-downloaded for the road ahead; there is no bundled emergency guidance; nothing tells the driver how stale the last information is.
Limited manager–driver coordination	A route change, a cancellation after pickup or a "stop and wait" instruction has no acknowledged, audited path from office to cab.
Insufficient route-specific safety intelligence	An "all clear" is shown when the system simply has no data. On a hill road, unknown is not safe.

1.3 Core Problem

Transport operators and drivers working across difficult, hazard-prone and connectivity-constrained corridors of the North Eastern Region need one platform that combines fleet and trip management, real road routing, evidence-based route risk assessment with a human authorisation step, near-live vehicle tracking, driver safety and emergency support, and a controlled offline mode — where every displayed judgement carries its evidence and its freshness, and where missing evidence is never presented as safety.

This statement is sharpened from the original brief by two findings in the project's own logs: the 2,908 km route incident of 14 September (a route was correct for its stops; the stops themselves were outside the region — docs/terrain/HANDOFF.md §12), and the hosted finding that with no landslide inventory configured every corridor is "review required", so the human authorisation step is not optional but mandatory (HANDOFF.md §16).

1.4 Stakeholder Pain Points

Fleet Manager

- Cannot tell whether a truck is moving, parked, or out of contact.
- Has to choose a road without evidence about its current condition.
- Cannot change a driver's instructions after pickup in a way that is confirmed and recorded.
- Is shown "safe" by tools that have no data.

Driver

- Loses navigation when data drops on the hill section.
- Receives official alerts (if at all) in a language or a form that does not say whether they apply to this road.
- Has no way to signal "I am fine, this is a rest stop" versus "I need help" other than a phone call that may not connect.
- Cannot read guidance written only in English.

Logistics Company

- No audit trail of who selected which route, on what evidence, and who authorised it.
- Duplicate or contradictory records when a manager double-clicks or two managers act at once.
- No single record of driver, truck, documents, assignment and trip history.

Emergency / Support Operations

- Learns about an incident late and second-hand.
- Has no frozen snapshot of where the truck was, what the last known conditions were, and whether the driver responded.

1.5 Why Existing Approaches Are Insufficient

The gap RASTA AI addresses is the *combination*: routing that is real (a routing engine, not a drawn line), risk that is evidence-based and honest about its gaps, a governance step where a named person accepts incomplete evidence for one dispatch, a driver application that degrades in a controlled way when the network goes, and a fleet console that shows the observed track against the planned corridor — all writing to one audit log. No single conventional tool does these together, and bolting them together without one authoritative backend recreates the coordination problem.

1.6 Design Constraints

Constraint	How the design responds (evidence)
Unreliable mobile data	Driver app polls; connection status = poll round-trip success; GPS fixes are queued durably on the phone and uploaded in batches (driver-app/src/tracking/tracker.ts, queueStorage.ts); an offline trip package is stored for 6 hours (driver-app/src/offline/packageStore.ts, PACKAGE_FRESH_MS).
GPS limitations	Accuracy-aware stationary filter; last-known position is shown with its age, never as live (driver-app/src/tracking/speed.ts, tracker.ts lastKnown); server judges staleness on server time.

Constraint	How the design responds (evidence)
API / provider failure	Every external provider sits behind an interface with a fallback and a timeout (routing 8 s, weather 6 s); an outage yields UNKNOWN/NOT_AVAILABLE, never a fabricated value (backend/app/core/config.py, backend/app/services/provider_health.py).
Weather-data freshness	Observation timestamps are carried; the driver's Safety card shows "N of M evidence factors available · updated ... ago".
Route-provider dependency	Primary URL configurable; public OSRM as keyless fallback; every returned route is validated against the requested endpoints before storage (backend/app/domain/routing.py, endpoint_mismatch).
Battery limitations	One GPS watcher; moving interval 10 s, stationary 60 s; batch size 6; trip poll runs 3× slower in background (backend/app/domain/telemetry_policy.py, TripProvider.tsx).
Offline operation	Bundled safety guide (en/hi/as), phrasebook, offline assistant intents, cached trip package; planned 9 km look-ahead prefetch is not on main (draft PR #1).
Hardware limitations	Android APK built with Expo; map on native rendered through a WebView (Leaflet); no proprietary SDK.
Security	Argon2id passwords, 15-minute access tokens, rotating refresh tokens with reuse detection, role permissions enforced server-side, rate limiting on login/refresh, RLS enabled on every table (docs/SECURITY.md, backend/app/core/).
Cost	Every data source is free/public: OSRM, Open-Meteo, MET Norway, OpenTopoData, GloFAS, NDMA SACHET, NASA GLC, Nominatim; hosting on Render free tier; Supabase free tier.
Scalability	Stateless FastAPI, connection pool sized for Supabase's session pooler (3 + 2 on Render), idempotent GPS ingestion, bounded polling (§11).

2. Proposed Solution

2.1 Solution Overview

RASTA AI is a two-client, one-backend logistics platform for the North Eastern Region. A **manager web console** plans shipments and trips, pairs drivers with trucks, asks the backend for real road routes, checks each candidate route against seven evidence sources (weather, terrain, landslide history, river discharge, official alerts, fleet traffic, routing metadata), routes any route with incomplete or elevated evidence to a named **authorised reviewer**, selects a route, dispatches the trip, and then follows the truck's observed GPS track against the planned corridor. A **driver application** (Android APK and a web build) receives the trip, verifies the truck with a photo, starts, navigates the pickup first and then the delivery with turn-by-turn guidance from the routing engine, uploads GPS in batches, raises a reroute request when it leaves the road, shows safety evidence and official alerts in the driver's language, keeps an offline emergency pack, and completes stops.

Everything consequential — assignment, route selection, reviewer authorisation, dispatch, reroute acceptance, cancellation with cargo disposition, emergency escalation — goes through the **FastAPI backend**, which is the only component that writes to the database. The backend's decisions are **deterministic rules over evidence**; two online language models are used only to word an explanation and never to decide. What the system cannot verify it labels UNKNOWN, and UNKNOWN is never rendered as safe.

2.2 Manager Platform

The manager console (manager-web/, React 19 + TypeScript, hosted at ner-manager.onrender.com) provides, each verified through the real UI on 16 September 2026 (HANDOFF.md §17):

Responsibility	What the manager does	Status
Fleet overview	Fleet page: quick actions (New trip, Assign truck, Review required, Active trips), "On the road" table with contact freshness (LIVE / stale / NO CONTACT) and stop progress, MapLibre map with observed track vs planned corridor.	IMPLEMENTED
Driver and truck records	Drivers page (create, profile drawer with licence health, documents, deactivate, "View as driver" read-only support session), Trucks page (create, retire, documents, verification state).	IMPLEMENTED
Driver assignment	One "Assign truck" dialog reachable from Fleet, Drivers, Trucks and the profile; server rules restated before the click (one-to-one pairing, live-trip guard, suspended/expired refusals); driver confirms the physical truck in the app.	IMPLEMENTED
Trip creation	Atomic shipment + trip planning (POST /api/trips/plan): client, cargo weight against truck capacity, pickup and destination as <i>confirmed locations</i> (address search, map pin or pasted Maps link), service-region check.	IMPLEMENTED
Route planning	POST /api/trips/{id}/routes/recalculate?detailed=true asks the routing engine for up to three options; distinct corridors are kept as EMERGENCY_BACKUP (max 2); each answer is validated against the requested endpoints.	IMPLEMENTED
Route review	"Check conditions & review" runs the recommendation engine; each candidate shows risk score/band, reason codes, evidence coverage (Weather / Terrain / Warnings / Traffic AVAILABLE or UNKNOWN), and its eligibility.	IMPLEMENTED
Route selection and dispatch	The control follows the server state: "Use this route" (eligible or authorised), "Open review" (review required), "Route blocked" (rejected). Dispatch is gated server-side on a selected, valid, authorised route.	IMPLEMENTED
Vehicle tracking	Fleet snapshot polled every 10 s; last position, speed, accuracy, reported age; track breadcrumb (gaps are drawn as gaps, not roads).	IMPLEMENTED (polling, not push)
In-trip control	Stop / change dialog for started trips: reason (≥ 10 characters) plus a cargo disposition after pickup; driver must acknowledge hold and route-change instructions.	IMPLEMENTED

Responsibility	What the manager does	Status
Reroute governance	Route options on a moving trip; driver's reroute proposal appears as a candidate; reviewer authorisation; "Reroute onto this".	IMPLEMENTED (inspector needs a reload to show a new proposal — open P3)
Emergency handling	Fleet Sentinel emergencies listed on the Fleet page with resolve / false-alarm.	IMPLEMENTED, NEEDS TESTING on hosted UI (sweep is on demand; scheduler disabled by default)
Diagnostics	System page: backend, database, provider health (routing, weather, terrain, flood, warnings, landslide inventory, AI providers, push, SMS) with truthful UNKNOWN / NOT_CONFIGURED states.	IMPLEMENTED

2.3 Driver Application

The driver application (driver-app/, Expo SDK 57 + React Native 0.86, APK 1.0.18; web build at ner-driver-web.onrender.com) provides:

Capability	Behaviour	Status
Sign-in	Phone number + password; session restore from secure storage.	IMPLEMENTED
Trip request and Accept	"NEW TRIP REQUEST" card; Accept records acceptance without starting the trip or sharing location.	IMPLEMENTED (physical phone, 16 Sep)
Truck verification	Photo (camera/gallery) + registration plate entered; backend records verification; mismatch is flagged to the manager.	IMPLEMENTED (physical phone, 14 Sep)
Start and stops	Start trip (needs a verified pairing), "Arrived at ..." / "Finish ..." per stop, pickup executes before delivery by construction, Complete trip.	IMPLEMENTED
Navigation	Navigate tab: next manoeuvre and distance from the routing engine's steps, remaining distance, route overview, terrain/landslide overlays, spoken guidance (device TTS). No maneuver is invented when the server sent none.	IMPLEMENTED
GPS tracking	Single watcher; accuracy-aware stationary filter; durable bounded queue; batched, idempotent upload; last-known position shown with its age.	IMPLEMENTED
Off-route and reroute	On-device projection with hysteresis (200 m enter / 80 m exit / 3 fixes); leaving the road pauses guidance and posts a reroute request; the new road is proposed to the manager, never applied automatically.	IMPLEMENTED
Safety information	Safety tab: emergency numbers (112 / 108 / 1033 hand-off to the dialler), bundled first-aid and road-safety guide (en/hi/as), danger cards with evidence classes (official alert, server decision, DEM, inventory).	IMPLEMENTED
Emergency tools	SOS with confirmation sheet; driver check-in ("I am safe / routine pause" vs "need help") feeding the Fleet Sentinel.	IMPLEMENTED; escalation path NEEDS TESTING end-to-end on hosted deployment
Roadside information	Nearby fuel, rest, repair and hospital from an OSM corridor snapshot (/me/trip/places).	IMPLEMENTED
Offline support	Trip package (selected route, backup route, stops, safety guide) cached for 6 h; assistant intents, phrasebook and guide work with the radio off.	PARTIALLY IMPLEMENTED (no look-ahead tile/route prefetch on main)
Safety assistant	Offline intent matching over application state in five languages; optional online explanation through the backend's AI proxy (wording only).	IMPLEMENTED
Connectivity awareness	"Connected / Last sync Ns ago" from poll success; isStale after a failed poll; never navigator.onLine.	IMPLEMENTED
Language	22 UI languages listed; English VERIFIED, several DRAFT, others FALLBACK_ENGLISH with the status shown to the driver (driver-app/src/i18n/appLanguage.ts).	PARTIALLY IMPLEMENTED

2.4 Backend Platform

The backend (backend/app, FastAPI on Python 3.11, hosted at ner-intelligence.onrender.com) is the system of authority.

Concern	Implementation	Evidence
Authentication	Local JWT: Argon2id password hashing, 15-minute access tokens, 30-day rotating refresh tokens with reuse detection (a replayed token revokes the whole family). Web clients receive the refresh token as an HttpOnly cookie; native clients in the body.	backend/app/services/auth.py, api/auth.py, docs/SECURITY.md §1
Authorization	Roles ADMIN / MANAGER / DRIVER / AUTHORISED_REVIEWER mapped to fine-grained permissions (route:select, route:review_authorize, trip:dispatch, fleet:location_read...); every mutating endpoint requires a permission; the reviewer deliberately lacks route:select.	backend/app/core/permissions.py, tests/test_authorization.py
API layer	14 routers, 84 route handlers under /api/*, Pydantic request/response models, consistent {error: {code, message, details}} envelope.	backend/app/api/*.py, docs/API_CONTRACTS.md
Trip lifecycle	DRAFT → ASSIGNED → (VERIFICATION_PENDING) → ACTIVE ⇌ DELAYED → DELIVERED → CLOSED, plus CANCELLED with post-pickup cargo dispositions; every transition audited.	backend/app/services/trips.py, domain/trip_state.py, models/enums.py
Routing	Provider chain (primary URL optional, OSRM fallback), 8 s timeout, up to 3 options, endpoint and detour validation, distinct-corridor de-duplication, EMERGENCY_BACKUP cap 2, planning never changes the current selection.	backend/app/services/routes.py, services/routing/osrm.py, domain/routing.py
Telemetry	POST /api/driver/me/location accepts batches; unique index on (trip_id, device_fix_id); device time and server time both stored; staleness judged on server time.	backend/app/services/telemetry.py, models/operations.py
Risk services	11-factor route risk (distance, duration, weather, landslide inventory, flood, historical incidents, elevation, official warnings, fuel model, road quality, truck restrictions — the last two always reported unavailable), fleet traffic reported alongside; bands LOW / MODERATE ≥ 30 / HIGH ≥ 60; unavailable factors listed; eligibility ELIGIBLE / REQUIRES_REVIEW / REJECTED / NOT_ASSESSED.	backend/app/domain/route_risk.py, route_eligibility.py
External APIs	Routing, weather, terrain, flood, warnings, geocoding, AI wording — all with timeouts and fallbacks; health reported on /api/system/providers.	backend/app/core/config.py, services/provider_health.py
Validation	Pydantic Coordinate bounds (PostGIS silently accepts swapped lat/lon), service-region bbox, capacity check before routing, reason length minimums, plausibility of reroute origins.	backend/app/schemas/common.py, schemas/domain.py, tests/test_geospatial.py
Error handling	Domain errors carry stable codes (ROUTE_SELECTION_REQUIRES_REVIEW, ROUTE_VALIDATION_FAILED, ASSIGNMENT_HAS_LIVE_TRIP, POST_PICKUP_RESOLUTION_REQUIRED...); provider outages are 503 ROUTING_UNAVAILABLE, refusals are 422; clients render the code's meaning before the click where possible.	backend/app/core/errors.py, docs/API_CONTRACTS.md
Database	Supabase PostgreSQL 17.6 + PostGIS 3.3 (ap-south-1), SQLAlchemy 2.0 async over psycopg3, 12 Alembic migrations, 20 tables.	backend/alembic/versions/, backend/app/models/

2.5 Intelligent / AI Layer

The project keeps a code-audited inventory of everything that could be called intelligence (docs/AI_INVENTORY.md, served live on /api/system/providers). The honest summary:

```

TRUE_LOCAL_ML           = 0      (no trained model is deployed)
LOCAL_LLM               = 0
TRUE_LOCAL_ML_EXPERIMENTAL = 1    (landslide-day logistic regression; research scripts only)
DETERMINISTIC_INTELLIGENCE = 20  (rules over evidence)
GEOMETRIC_ALGORITHM     = 5      (projection, gradient, next-turn, speed filter)
OFFLINE_KNOWLEDGE_SYSTEM = 4      (phrasebook, safety guide, reason codes, places snapshot)
ONLINE_LLM              = 2      (Gemini primary, OpenRouter fallback – wording only)
PROVIDER_MODEL_OUTPUT    = 5      (weather, flood and DEM providers, routing engine)

```

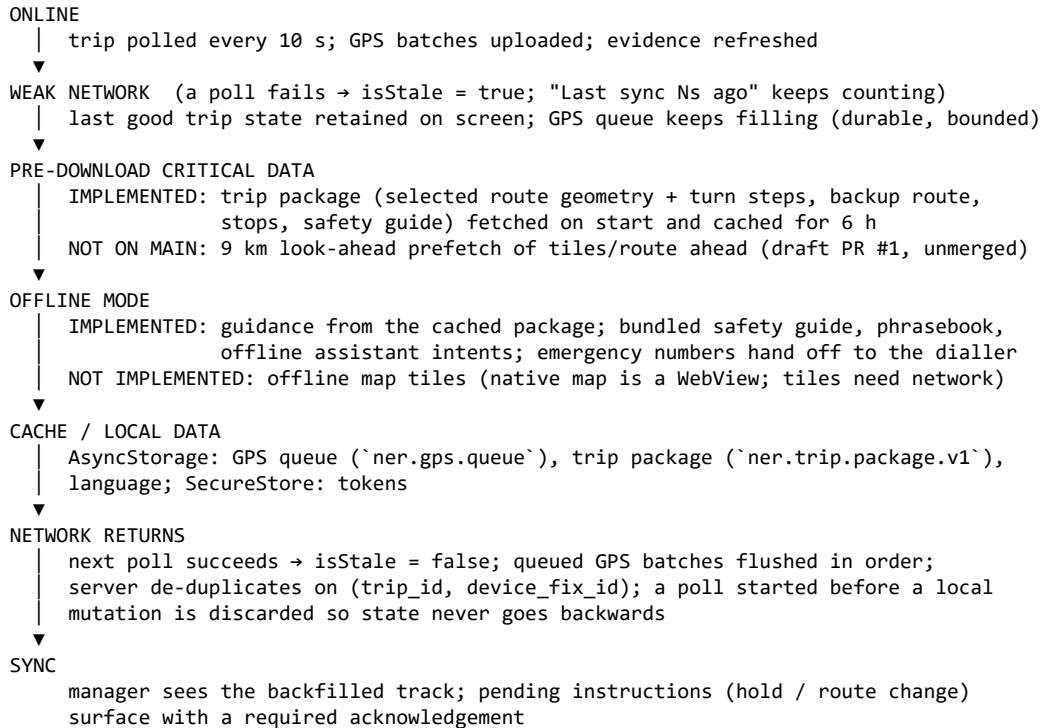
Deterministic logic (decides): route risk scoring, eligibility refusal, reroute governance, route recommendation, Fleet Sentinel escalation, off-route projection, telemetry policy, capacity and pairing invariants, dispatch gate. All are plain functions with published thresholds and reason codes (backend/app/domain/*.py).

AI / ML functionality (does not decide): two online language models produce a worded explanation of facts the backend already computed; the prompt carries the facts and the model may not add any (backend/app/services/gemini.py, api/ai.py). An Ollama client exists but has never had a model installed. The landslide model is **BLOCKED_BY_DATA**: the public inventory ends in 2017 and is too sparse for a calibrated

probability (docs/AI_MODELS.md §0a). The fuel estimate is a physics-informed baseline with hand-set constants, labelled advisory, not a trained model.

The phrase the team uses with judges is *evidence-based decision support with deterministic rules; language models word the answers and never decide*.

2.6 Offline / Connectivity-Aware Strategy



An SMS fallback path is **NOT CONFIGURED** (SMS_PROVIDER unset; Diagnostics shows the row as such). Push notifications are relayed server-side through Expo, but the installed APK was built without `google-services.json`, so device push is **BLOCKED**; the app relies on polling plus local notifications.

3. Unique Key Features

Each feature below is described the same way. "Current implementation status" uses the six labels defined at the top of the document, with the evidence that supports the label.

3.1 Validated Route Planning (real roads, checked answers)

- **Purpose:** Give the manager real road routes for a trip, never a drawn line.
- **Problem solved:** Consumer routers return whatever connects two points; a wrong stop or a snapped endpoint produced a confident 2,908 km "route" on 14 Sep.
- **How it works:** Backend asks the routing engine for up to three options with turn instructions; each answer is checked to start within 10 km of the pickup, end within 10 km of the destination, be no shorter than the straight line and no longer than $3 \times \text{straight line} + 20 \text{ km}$; distinct corridors are kept as EMERGENCY_BACKUP (max 2); the trip's current selection is never changed by planning.
- **Inputs:** confirmed pickup and destination coordinates (in the service region), routing profile.
- **Processing:** `routes.plan()` → provider chain → `endpoint_mismatch` → `is_distinct_corridor` → persist PROPOSED routes, supersede obsolete unselected ones.
- **Output:** `trip_routes` rows with geometry, distance, free-flow duration, maneuvers, provider name.
- **User:** Manager (planning), Driver (drives the selected one).
- **Offline capability:** None at planning time (server-side); the selected route is cached on the phone.
- **Current implementation status:** **IMPLEMENTED** — `backend/app/services/routes.py`, `domain/routing.py`; tests `test_route_api.py`, `test_corridor_too_wide.py`, `test_routing.py`; hosted planning 2.9 s on 16 Sep.
- **Technical dependency:** OSRM (public fallback or configured primary), PostGIS geography.
- **Failure behaviour:** provider down → 503 ROUTING_UNAVAILABLE (retry sensible); no road → 422 NO_VIABLE_ROUTE; answer does not match the question → 422 ROUTE_VALIDATION_FAILED, nothing stored.
- **Potential future enhancement:** self-hosted Valhalla/OSRM with a truck profile (height/weight restrictions); the interface already exists (`services/routing/base.py`).

3.2 Planned Route vs Observed GPS Track

- **Purpose:** Let a manager see where the truck actually went against where it was told to go.
- **Problem solved:** A planned corridor alone hides departures from the road and GPS gaps.
- **How it works:** Fleet map draws the planned polyline and the observed breadcrumb separately; gaps in GPS are drawn as gaps ("A gap is not a road"); the inspector shows last position, speed, accuracy and reported age.
- **Inputs:** `trip_routes.geometry`, `gps_points` (recent track endpoint).
- **Processing:** GET `/api/trips/{id}/track`, GET `/api/fleet/active` polled every 10 s.
- **Output:** two layers on MapLibre; LIVE / stale / NO CONTACT badge by reported age.
- **User:** Manager.
- **Offline capability:** Not applicable (console).
- **Current implementation status:** **IMPLEMENTED** — `manager-web/src/components/FleetMap.tsx`, `components/track.ts`, `pages/FleetPage.tsx`; hosted, 16 Sep.
- **Technical dependency:** MapLibre GL, OSM raster tiles.
- **Failure behaviour:** a failed poll shows the last good snapshot with a stale banner; it never blanks the map.
- **Potential future enhancement:** server push (WebSocket) instead of 10 s polling.

3.3 Real Route Alternatives (no fabricated backups)

- **Purpose:** Offer the manager a genuine second corridor when one exists, and say plainly when it does not.

- **Problem solved:** Earlier UI implied three routes always existed; on most NER corridors there is one sensible road.
- **How it works:** Extra provider options are stored only when they are a distinct corridor; the panel reads "1 corridor offered — that is the answer, not a shortfall" when so.
- **Inputs / Processing / Output:** as 3.1; cards per route with state SELECTED / SELECTABLE / REVIEW REQUIRED / BLOCKED / NOT CHECKED / STALE.
- **User:** Manager.
- **Current implementation status: IMPLEMENTED** —
manager-web/src/components/RouteCandidateCards.tsx; MAX_EMERGENCY_BACKUPS = 2; test
test_route_api.py cap test.
- **Failure behaviour:** none needed; absence of alternatives is a valid state.
- **Potential future enhancement:** manager-defined via-points.

3.4 Evidence-Based Route Risk (UNKNOWN ≠ SAFE)

- **Purpose:** Score each candidate route from evidence and state which evidence was missing.
- **Problem solved:** Tools that show "clear" when they have no data.
- **How it works:** route_risk.assess scores eleven named factors — distance, duration, weather (rain intensity and coverage at up to 10 route points, wind gusts), landslide inventory, flood (river discharge vs 30-day mean), historical incidents (recorded slides within 5 km), elevation (terrain gradient), official warnings (CAP alerts matched by district), fuel model, road quality and truck restrictions (the last two always reported as unavailable) — and reports fleet traffic from own probes alongside; score → band LOW < 30 ≤ MODERATE < 60 ≤ HIGH; every unavailable factor is listed.
- **Inputs:** route geometry, provider observations, static inventories.
- **Output:** score, band, reason codes (translated en/hi/as), per-factor availability, evidence coverage line.
- **User:** Manager (review), Driver (Safety cards, Personal Route AI explanation).
- **Offline capability:** Driver keeps the last assessment with its timestamp in the trip package.
- **Current implementation status: IMPLEMENTED** — backend/app/domain/route_risk.py (+ weather.py, terrain.py, landslide.py, flood.py, warnings.py, traffic.py); tests test_route_risk.py, test_landslide_route_risk.py, test_flood_context.py, test_official_warnings.py, test_terrain_route_risk.py. Hosted cold check 9.3 s, warm 5.3 s (16 Sep).
- **Technical dependency:** Open-Meteo / MET Norway, Open-Meteo elevation / OpenTopoData, GloFAS, NDMA SACHET, NASA GLC snapshot.
- **Failure behaviour:** a provider that does not answer marks its factor NOT_AVAILABLE; the band still computes from what is available and the coverage line says what is missing.
- **Potential future enhancement:** a current landslide inventory (state DDMA feeds) would unblock the landslide model.

3.5 Route Governance: Review Authorisation and the Dispatch Gate

- **Purpose:** A route with incomplete or elevated evidence can be used only after a named reviewer accepts responsibility for one dispatch.
- **Problem solved:** "Click through the warning" culture; silent acceptance of risk.
- **How it works:** Eligibility REQUIRES_REVIEW blocks selection; an AUTHORISED_REVIEWER (who cannot select routes) records a rationale (≥ 20 characters) bound to the evidence digest; the authorisation is spent inside the selection transaction and expires; a hazard-REJECTED route cannot be authorised by anyone. Dispatch is refused server-side without a selected, valid route.
- **Inputs:** assessment digest, reviewer identity, rationale.
- **Output:** route_review_authorizations row; audit entry; selection.

- **User:** Reviewer, Manager.
- **Current implementation status: IMPLEMENTED** — migration 0007, backend/app/services/route_review.py, routes.apply_selection, trips._assert_dispatchable_route; tests test_route_review_authorization.py, test_dispatch_route_gate.py, test_route_selection_hazard_api.py; hosted end to end 16 Sep. On the hosted deployment every corridor is REQUIRES_REVIEW (no landslide inventory configured), so this path runs on every trip.
- **Failure behaviour:** the UI names the way forward ("Open review") instead of a dead button; server refusals carry ROUTE_SELECTION_REQUIRES_REVIEW, ROUTE_REJECTED_ACTIVE_HAZARD, ROUTE_ELIGIBILITY_NOT_ASSESSED.
- **Potential future enhancement:** two-person control for HIGH band.

3.6 Pickup-First Navigation with Turn Guidance

- **Purpose:** The driver's first target is the pickup; guidance switches to the delivery once the pickup is completed.
- **How it works:** Stops execute in sequence; the Navigate tab reads the routing engine's step list and selects the next manoeuvre from the driver's projected position; distance to next turn and remaining distance are shown; spoken guidance uses device text-to-speech.
- **Inputs:** selected route maneuvers (migration 0009), GPS fixes.
- **User:** Driver.
- **Offline capability:** guidance continues from the cached package; map tiles need network.
- **Current implementation status: IMPLEMENTED** — driver-app/src/map/maneuvers.ts, navState.ts, NextTurnPanel.tsx, useSpokenGuidance.ts; physical phone: "1 Pickup NEXT", "Arrived at Pickup" first, delivery next after completion (16 Sep).
- **Failure behaviour:** no maneuvers from the server → "Guidance unavailable", never an invented instruction.
- **Potential future enhancement:** lane guidance; offline tiles.

3.7 Off-Route Detection and Governed Reroute

- **Purpose:** Detect a truck that has left the planned road and get it a real new road, approved by a person.
- **How it works:** Server projection with a 200 m threshold and on-device projection with hysteresis (200/80 m, 3 fixes); the phone pauses guidance, shows "Off the planned road · new road N km awaits manager", and posts POST /me/trip/reroute with its position; the backend plans a real road from there (validated against the planned road's length), stores it as EMERGENCY_BACKUP; the manager's route options show it; reviewer authorises; "Reroute onto this" applies it; the old route stays as history; the phone follows the new road on its next poll.
- **Current implementation status: IMPLEMENTED** — backend/app/domain/route_progress.py, services/reroute.py, api/driver.py reroute; tests test_reroute.py, test_reroute_api.py, test_driver_reroute_api.py; hosted: off-route at 22 km → proposal 76.6 km → authorised → accepted → phone on the new road (16 Sep).
- **Failure behaviour:** a position far outside the corridor is refused (422) so a 2,451 km "backup" can no longer be stored (fixed 16 Sep, commit 7b56554).
- **Potential future enhancement:** manager notification when a proposal arrives (today the inspector needs a reload — open P3).

3.8 Connectivity-Aware Driver App and GPS Queue

- **Purpose:** Keep tracking honest and cheap on a bad network.
- **How it works:** connection state = last poll succeeded; one GPS watcher; moving 10 s / stationary 60 s; fixes queued in a durable bounded queue and uploaded in batches of 6 (max 500); server idempotent on (trip_id, device_fix_id); last-known position shown with age.

- **Current implementation status: IMPLEMENTED** — driver-app/src/tracking/*, backend/app/domain/telemetry_policy.py, services/telemetry.py; tests tracker.test.ts, queueStorage.test.ts, speed.test.ts, test_telemetry.py.
- **Failure behaviour:** 422 on a malformed batch drops that batch permanently; other failures back off and retry; queue overflow keeps the newest fixes.
- **Known gap:** with device location switched off the app still uploads a cached fix labelled "GPS live" (open P3, HANDOFF.md §17).

3.9 Pre-download / Offline Trip Package

- **Purpose:** Carry the trip's road, backup road, stops and safety guide through a dead zone.
- **How it works:** GET /api/driver/me/trip/offline-package at start; stored in AsyncStorage under ner.trip.package.v1; CURRENT for 6 h, then STALE (still shown, labelled).
- **Current implementation status: PARTIALLY IMPLEMENTED** — package: backend/app/services/offline_package.py, driver-app/src/offline/packageStore.ts, tests test_offline_package.py, packageStore.test.ts. Look-ahead prefetch of 9 km ahead: **NOT IMPLEMENTED on main** (draft PR #1, unmerged). Offline map tiles: **NOT IMPLEMENTED**.
- **Failure behaviour:** package fetch failure → app continues online-only and says so.

3.10 Weather, Landslide and Disruption Awareness on the Phone

- **Purpose:** Tell the driver what is ahead, from which source, and how fresh it is.
- **How it works:** Danger cards carry an evidence class (official alert / server decision / DEM / inventory), a distance ahead, and a "N of M evidence factors available · updated ... ago" line; official NDMA alerts are matched to corridor districts; cards de-duplicate with a cooldown.
- **Current implementation status: IMPLEMENTED** — driver-app/src/safety/riskCards.ts, notify/local.ts, backend/app/domain/warnings.py; physical phone showed a CRITICAL IMD thunderstorm alert for Assam districts on 16 Sep.
- **Failure behaviour:** feed unavailable → "NOT CHECKED", never "clear".

3.11 Driver Safety Assistant (offline first, online wording optional)

- **Purpose:** Answer "what do I do now" in the driver's language without a network.
- **How it works:** keyword intent matching over application state (trip, route risk, breaks, guide) in five languages; assistant.test.ts asserts it imports no network client; an optional online answer is worded by Gemini/OpenRouter through POST /api/ai/ask with the facts supplied by the backend.
- **Current implementation status: IMPLEMENTED** offline path; online wording **IMPLEMENTED** but depends on GEMINI_API_KEY / OPENROUTER_API_KEY being set on the host (both reported UNKNOWN on Diagnostics at the time of writing → effectively **BLOCKED** on hosted until keys are set).

3.12 Emergency Assistance and Fleet Sentinel

- **Purpose:** Turn "stationary and silent" into a checked, escalated, recorded event.
- **How it works:** Sentinel rules: stationary ≥ 60 min within 100 m outside a stop, or no contact ≥ 60 min → driver check-in required; "I am safe" clears; "need help" or an expired deadline → SOS escalated with a frozen briefing snapshot; manager resolves or marks false alarm; driver-side SOS opens a confirmation sheet and hands off to the dialler (112 / 108 / 1033).
- **Current implementation status: IMPLEMENTED** — backend/app/domain/sentinel.py, services/sentinel.py, api/emergencies.py, migration 0010, driver-app/src/screens/SafetyScreen.tsx, TripScreen.tsx check-in; tests test_sentinel.py, test_sentinel_concurrency.py, test_emergency_api.py, test_driver_decision.py. **NEEDS TESTING** end-to-end on the hosted deployment; the periodic sweep scheduler is disabled by default (SENTINEL_SCHEDULER_ENABLED = False), so sweeps run on demand.

3.13 Roadside Facilities

- **How it works:** GET /api/driver/me/trip/places returns fuel, rest, repair and medical points near the corridor from an OSM snapshot bundled with the backend; Google Places is an optional key.
- **Current implementation status: IMPLEMENTED** — backend/app/services/places/snapshot.py, test test_places_snapshot.py; driver Navigate → "Find a place to stop".

3.14 Trip Lifecycle with Post-Pickup Cargo Rules

- **How it works:** A trip cannot be cancelled silently once cargo is on the truck: the manager must give a reason (≥ 10 characters) and a disposition (RETURN_TO_DEPOT, NEW_DESTINATION, HOLD_FOR_INSTRUCTION, COMPLETE_CURRENT_LEG, CARGO_UNLOADED); hold and route-change instructions require the driver's acknowledgement; every step is a trip_events row.
- **Current implementation status: IMPLEMENTED** — backend/app/services/trips.py DISPOSITIONS, driver_trips.acknowledge_instruction; tests test_post_pickup_resolution.py (10 cases); hosted smoke 16 Sep.

3.15 Fleet Monitoring and Manager–Driver Coordination

- **Current implementation status: IMPLEMENTED** as near-live polling (10 s manager, 10 s driver, 3× slower in background). Not real-time push. Measured cross-device latencies on 16 Sep: dispatch → phone ≤ 9 s; start → manager 10 s; delivery → manager ≤ 45 s including the driver's taps.

3.16 Battery / Last-Known-Location Strategy

- **Current implementation status: IMPLEMENTED** — quantised last-known age, stationary interval 60 s, single watcher, no network call per fix (batched), background poll slowdown.

3.17 Multilingual Assistance

- **Current implementation status: PARTIALLY IMPLEMENTED** — safety guide and reason codes reviewed in en/hi/as; UI strings for 22 listed languages with a shown status (English VERIFIED; Assamese, Bengali, Hindi, Gujarati, Kannada... DRAFT; Bodo, Dogri, Kashmiri, Konkani... FALLBACK_ENGLISH); phrasebook bundled; device speech recognition/TTS.

3.18 One-to-One Driver–Truck Assignment with Physical Verification

- **Current implementation status: IMPLEMENTED** — partial unique indexes (migration 0006) guarantee one live pairing per driver and per truck; a pairing with a trip under way cannot be broken (ASSIGNMENT_HAS_LIVE_TRIP); the driver verifies the physical truck with a photo and plate before the first start; manager can verify manually. Hosted UI certified 16 Sep.

4. Target Users

4.1 Fleet Manager (dispatcher)

- **Needs:** Know where every truck is and how fresh that knowledge is; choose a road on evidence; change instructions with a record; never dispatch blind.
- **Responsibilities:** Create trips, pair drivers with trucks, plan and check routes, send routes for review, dispatch, watch the fleet, respond to holds and emergencies, close delivered trips.
- **System interaction:** Manager web console (Fleet, Trips, Drivers, Trucks, Review, Diagnostics).
- **Primary benefits:** One screen per decision; refusals explained before the click; observed track vs plan; audit trail.

USER	Fleet manager
↓	
PROBLEM	A truck is out of contact on a hill road; the road ahead may be closed
↓	
SYSTEM	Fleet page shows NO CONTACT with age; Route tab checks conditions (rain, alerts, landslide history, river); reroute options are real roads
↓	
RESULT	A recorded decision: hold, reroute (after review), or continue – with evidence

4.2 Truck Driver

- **Needs:** Know the next turn and the pickup first; be told about hazards in a language they read; keep working when data drops; get help fast.
- **Responsibilities:** Accept, verify the truck, start, follow the road, complete stops, acknowledge instructions, check in when asked.
- **System interaction:** Driver app (Trip, Navigate, Safety, More) on an Android phone; web build as fallback.
- **Primary benefits:** Turn guidance and safety evidence in one app; no dead ends offline; SOS that cannot dial by accident.

USER	Driver
↓	
PROBLEM	Data drops on the ghat section; an IMD thunderstorm alert covers the district ahead
↓	
SYSTEM	Cached trip package keeps guidance; danger card names the source and its age; Safety tab works with the radio off; GPS fixes queue and upload later
↓	
RESULT	The driver keeps navigating, knows what the alert means for this road, and the manager sees the backfilled track when the network returns

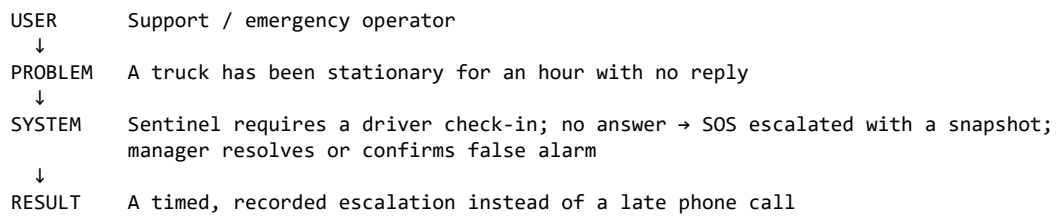
4.3 Logistics Company / Fleet Operator

- **Needs:** Accountability and continuity: who decided what, on what evidence; no duplicate or lost records; a system that outlives a WhatsApp group.
- **Business value:** Fewer stranded loads; documented risk acceptance; driver and vehicle records with document expiry tracking.
- **Operational value:** One backend enforcing capacity, pairing and dispatch rules; audit log on every consequential change (audit_logs).

USER	Operator
↓	
PROBLEM	A consignment was lost to a landslide; nobody can show what was known and who approved the road
↓	
SYSTEM	Route review authorisation binds a reviewer, a rationale and the evidence digest to one selection; audit_logs holds every transition
↓	
RESULT	A defensible record and a repeatable process

4.4 Emergency / Support Personnel

Included because the design supports it: the Fleet Sentinel escalation produces a frozen briefing snapshot (position, last conditions, driver response) and the manager's "View as driver" read-only support session lets a support operator see the driver's screen state without the driver's password.



5. System Architecture

5.1 High-Level Architecture

The diagram below is drawn from the repository as it is, not from the earlier design (docs/ARCHITECTURE.md shows a WebSocket channel and an ML service; neither exists in code — clients poll, and no trained model is deployed).

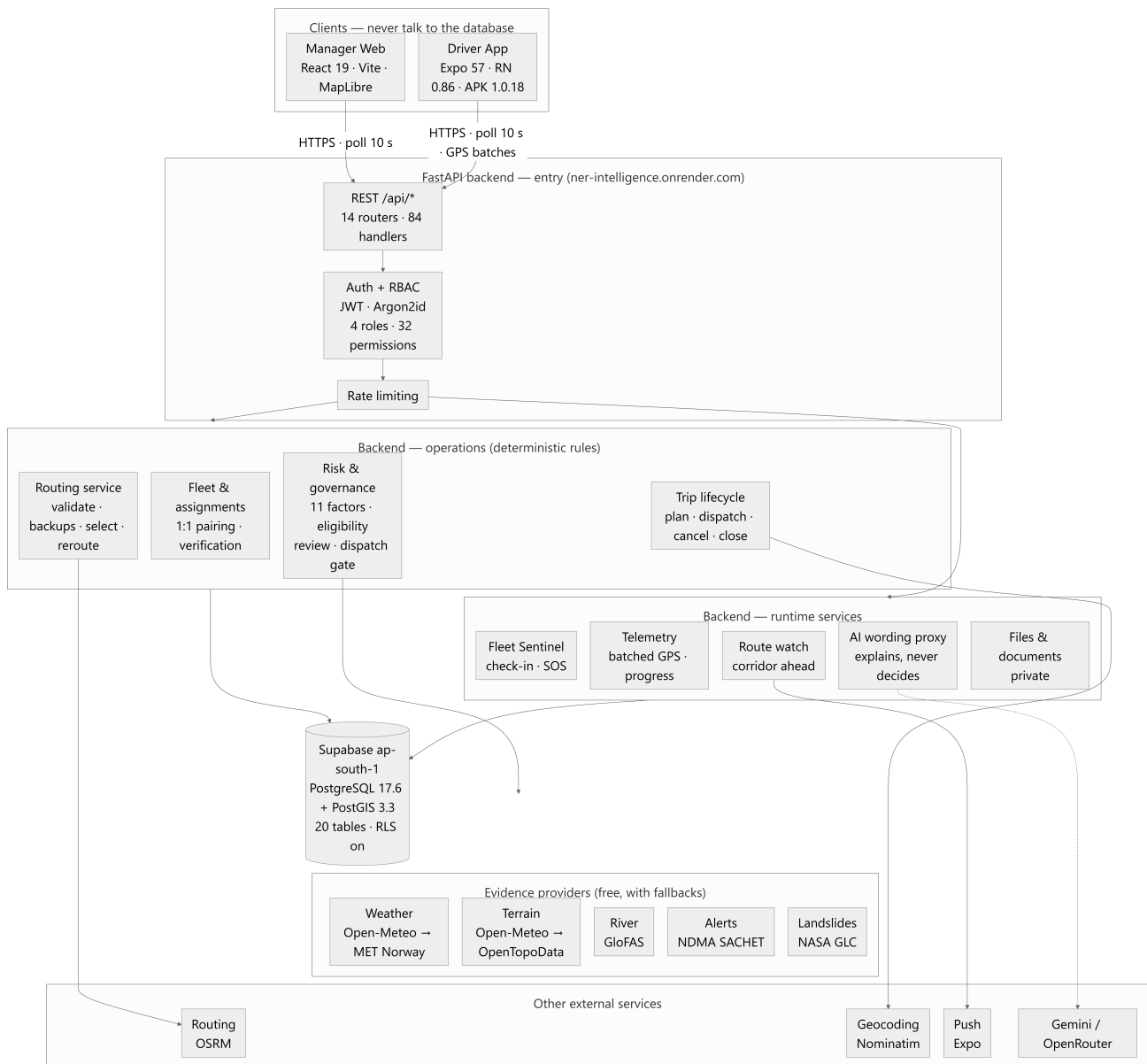


Figure 1 — High-level architecture (rendered from the Mermaid source in the Markdown edition)

5.2 Client Layer

Manager web application. A single-page React 19 application built with Vite, styled with Tailwind 4, routed with React Router 7. Pages: Login, Fleet (map + on-the-road table + inspector with Overview / Route / Cargo / Activity tabs), Trips (planner + trip review + trips table), Drivers, Trucks, Assignment records, Review (reviewer role), Diagnostics. The API client (manager-web/src/api/client.ts) wraps fetch with a 15 s timeout (90 s for the two evidence fan-out reads), one silent token refresh on 401, and a typed ApiError. Polling hooks (useFleetPoll, useResource) retain the last good reading on failure. No business rule is decided in the client; the client restates server rules to prevent dead clicks.

Driver application. Expo SDK 57 / React Native 0.86 with TypeScript. Tabs: Trip, Navigate, Safety, More (My details, Assistant, Language, Theme). The map is Leaflet in a WebView on native and Leaflet directly on web (DriverRouteMap.native.tsx / .web.tsx), with static and live layers drawn in separate passes. TripProvider polls the current trip every 10 s and is the single source of connection truth. useLocationTracking owns the one GPS watcher and the durable queue.

Map / navigation UI. Manager: MapLibre GL with OSM raster tiles, planned corridor vs observed track, terrain and hazard overlays. Driver: Leaflet with route polyline, next-turn panel, terrain/landslide overlays, route overview, re-centre.

Local cache (driver). AsyncStorage: GPS queue, trip package (6 h), language choice. SecureStore: access and refresh tokens. No tile cache.

5.3 Backend Layer

Element	Responsibility	Where
API routers	/api/auth, /api/drivers, /api/trucks, /api/assignments, /api/driver/me/*, /api/shipments, /api/trips/*, /api/fleet, /api/emergencies, /api/geocoding, /api/ai, /api/files, driver/manager documents, /health, /ready, /api/system/*	backend/app/api/
Services	Orchestration and transactions: trips, routes, route_review, reroute, route_recommendation, route_risk, telemetry, driver_trips, assignments, drivers, trucks, shipments, sentinel, notify, offline_package, navigation, provider_health, simulation (demo only)	backend/app/services/
Domain (pure rules)	route_risk, route_eligibility, route_recommendation, reroute, routing (validation), route_progress, terrain, weather, flood, landslide, warnings, traffic, sentinel, telemetry_policy, trip_state, fuel_model, road_memory	backend/app/domain/
Authentication	Argon2id, HS256 JWT signed with SECRET_KEY, 15 min access, 30 day rotating refresh with family revocation on reuse, cookie for web / body for native	backend/app/services/auth.py, core/security.py
Authorization	require_permission(...) dependency on every mutating handler; role → permission map	backend/app/core/permissions.py, api/deps.py
Routing abstraction	RoutingProvider interface, RoutingChain primary → fallback, OSRM implementation	backend/app/services/routing/
Trip management	State machine, dispatch gate, cancellation rules, instructions with acknowledgement	backend/app/services/trips.py, domain/trip_state.py
Telemetry	Batch ingest, idempotency, server-time staleness, route progress projection	backend/app/services/telemetry.py, domain/route_progress.py
Risk calculation	11-factor assessment, eligibility, recommendation comparison rule	backend/app/domain/route_risk.py et al.
Provider integration	Weather (2), terrain (2), flood, warnings, landslide snapshot, geocoding, AI wording, push	backend/app/services/*
Background workers	Route-ahead watch (ROUTE_WATCH_ENABLED, 60 s tick), warnings poller, Sentinel scheduler (off by default)	backend/app/services/route_watch.py, warnings.py, sentinel.py

5.4 Data Layer

Database: Supabase PostgreSQL 17.6 with PostGIS 3.3 in ap-south-1. Schema managed by 12 Alembic migrations (0001_bootstrap_postgis ... 0012_push_notifications). All geometry is geography(..., 4326) so distances are metres. UUID primary keys, TIMESTAMPTZ everywhere, soft delete on master data, native enums.

Main entities (20 tables):

Group	Tables
Identity	users, refresh_tokens, drivers, driver_documents
Fleet	trucks, truck_documents, truck_maintenance, driver_truck_assignments
Operations	shipments, cargo_items, trips, trip_stops, trip_routes, trip_events, driver_notifications
Telemetry	gps_points (unique (trip_id, device_fix_id))
Governance & safety	route_review_authorizations, emergencies, audit_logs

Group	Tables
Files	stored_files

Relationships: a user may be a driver; a driver holds at most one live assignment to one truck (partial unique indexes); a shipment has cargo items and one trip; a trip has ordered stops, many routes (one SELECTED / current), events, GPS points, review authorisations and at most one active emergency.

Persistence rules: every consequential mutation writes audit_logs (actor, action, before/after); route history is superseded, never deleted; GPS accepts late and batched data.

RLS / authorization: Row Level Security is enabled on every table with no policies, so the Supabase Data API exposes nothing with the anon key; the backend connects as the database role and is the only writer. Authorization is enforced in the backend, not in RLS (docs/SECURITY.md §5).

Design-only entities (NOT IMPLEMENTED): payments, expenses, payroll, deliveries (proof of delivery), road_incidents, weather_events (docs/DATA_MODEL.md).

5.5 External Services

Service	Purpose	Data	Fallback	Failure handling
OSRM (public router.project-osrm.org; primary URL configurable)	Road routing with turn steps	Geometry, distance, duration, maneuvers	Primary → public OSRM	8 s timeout; 503 ROUTING_UNAVAILABLE; 422 on no route or failed validation; nothing stored
Open-Meteo	Weather at up to 10 route points	Precipitation, wind gusts, timestamps	MET Norway locationforecast	6 s timeout; factor NOT_AVAILABLE; band computed from the rest
Open-Meteo elevation (Copernicus DEM)	Terrain profile	Elevation samples → gradient, steep km	OpenTopoData (SRTM)	Coverage below floor → "PARTIAL", not scored
Open-Meteo flood API (GloFAS)	River discharge context	Discharge vs 30-day mean at corridor cells	None	"NOT MEASURED"; never a closure claim
NDMA SACHET CAP RSS	Official disaster alerts	Event, severity, districts, expiry	None (10 min TTL cache)	"NOT CHECKED"; alerts placed by district name only
NASA Global Landslide Catalog	Historical landslide exposure	Bundled 2007–2017 snapshot	None	Inventory absent → LANDSLIDE_DATA_NOT_CONFIGURED → REQUIRES_REVIEW
Nominatim (OSM)	Address search; Google Maps link resolver	Coordinates, labels	None (map pin, paste link)	"Address search temporarily unavailable"; manager can choose on map
Gemini Developer API	Wording of explanations	Text	OpenRouter free models	Deterministic text remains; AI status shown on Diagnostics
Expo push service	Push alerts to phones	Tokens, messages	Local notifications + polling	Relay UNKNOWN without google-services.json in APK
Own fleet probes	Traffic	Observed vs planned pace, 15 min window	—	Fewer than two vehicles → UNKNOWN, never CLEAR

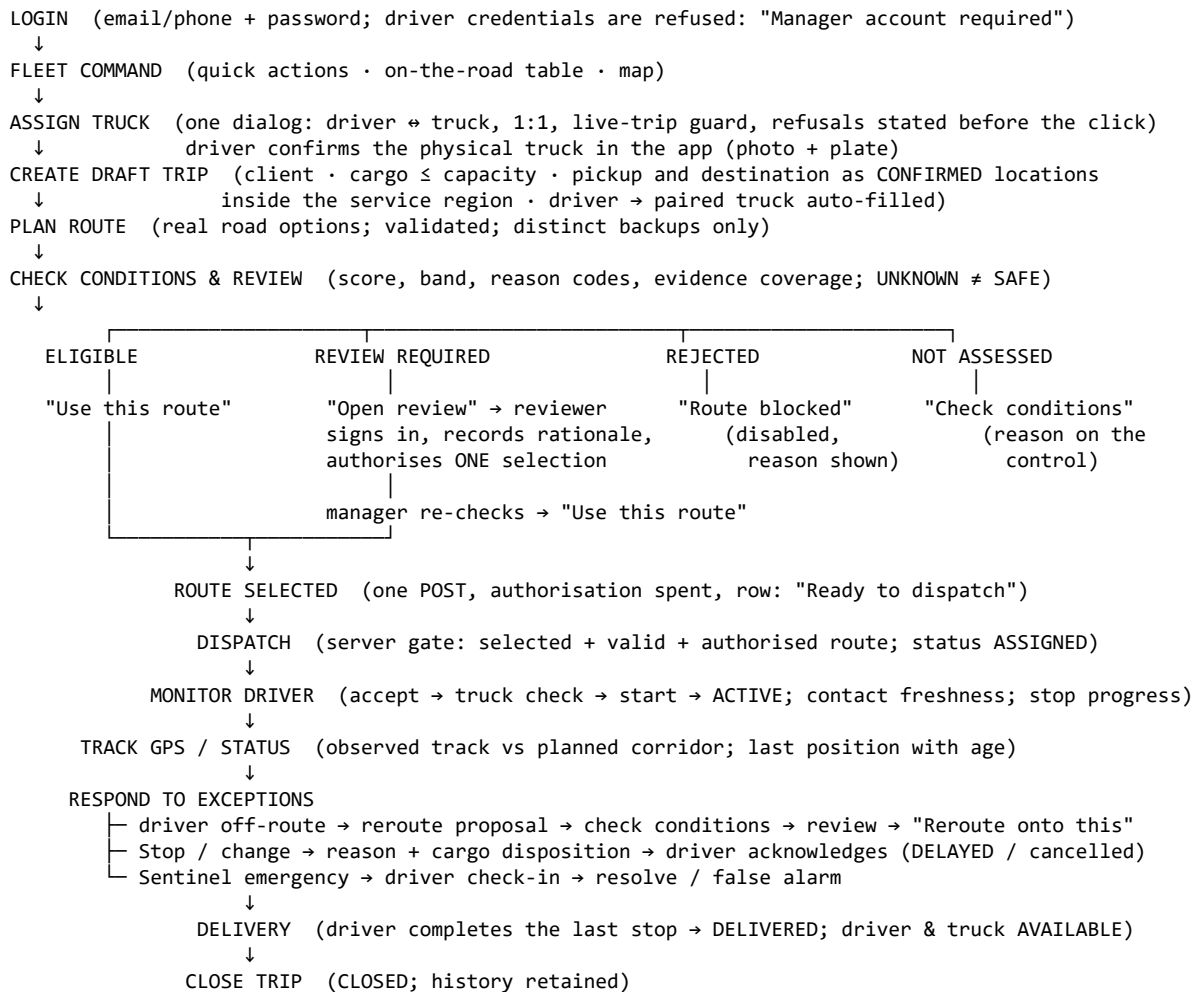
Not present and not claimed: paid traffic APIs, satellite imagery, SMS gateway (SMS_PROVIDER unset → NOT_CONFIGURED).

5.6 Architecture Principles

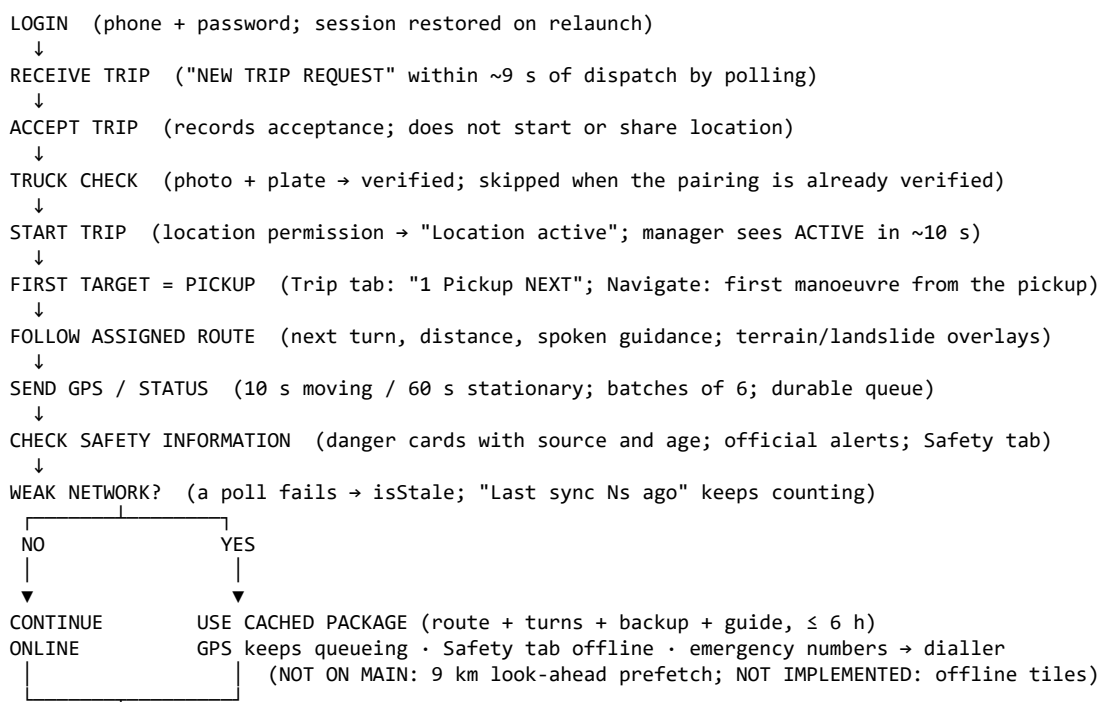
- **Separation of concerns:** pure domain rules (domain/) are importable without a database; services own transactions; routers own HTTP.
- **Provider abstraction:** every external source sits behind an interface with a timeout and a fallback; a swap is configuration.
- **Deterministic decision logic:** decisions are functions with published thresholds and reason codes; language models word, never decide.
- **API validation:** Pydantic models with coordinate bounds, region bbox, reason-length and disposition validation; capacity checked before routing.
- **Authentication and authorization:** short-lived tokens, rotating refresh with reuse detection, role→permission map enforced per handler; reviewer cannot select, driver cannot use the manager console.
- **Failure handling:** stable error codes; provider outages are 503, refusals 422; clients render the state before the click.
- **Offline degradation:** poll-based truth, durable GPS queue, cached trip package, bundled guidance.
- **Data freshness:** every reading carries its age; staleness judged on server time; UNKNOWN is a first-class state.
- **Security:** no client holds a database credential; RLS on; private files; rate limits; audit log.
- **Scalability:** stateless API, idempotent ingestion, bounded polling, pooled connections (\$11).

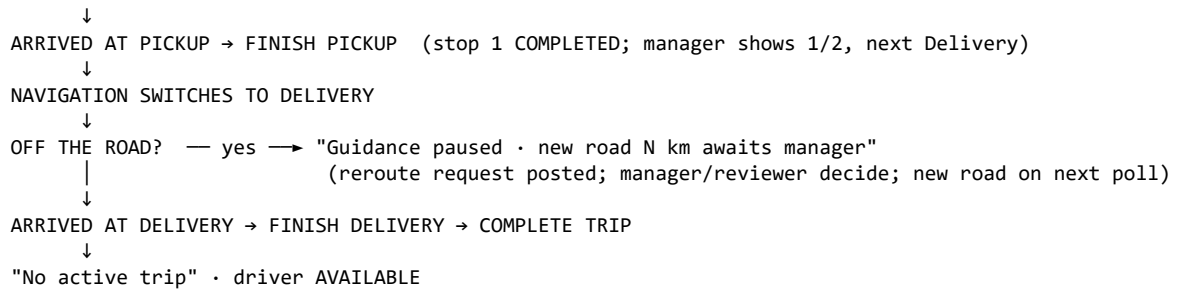
6. Workflow / User Flow Diagram

6.1 Manager Workflow (as the hosted console behaves, certified 16 Sep 2026)

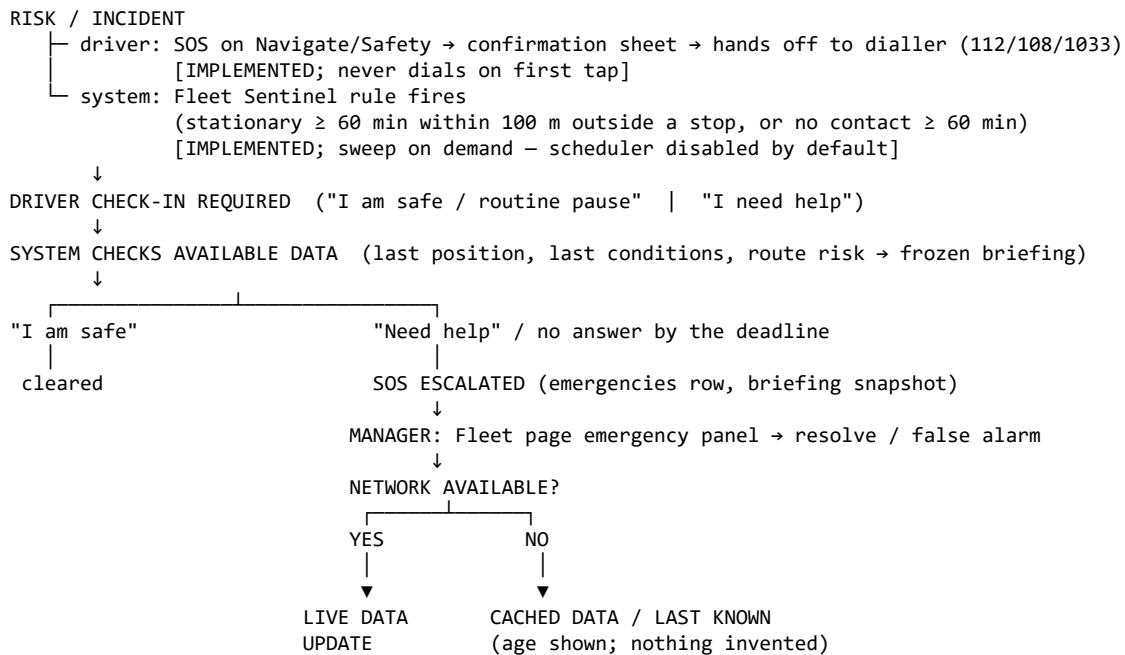


6.2 Driver Workflow (physical phone, APK 1.0.18, certified 16 Sep 2026)





6.3 Emergency Flow



Implemented vs planned: SOS hand-off, check-in, Sentinel rules, escalation, manager resolution and the frozen snapshot are in code with tests; the hosted end-to-end run of an escalation is **NEEDS TESTING**; automatic periodic sweeps are **BLOCKED** by configuration (SENTINEL_SCHEDULER_ENABLED=False); SMS to a manager is **NOT CONFIGURED**; push to the phone is **BLOCKED** (APK without google-services.json).

7. Data Flow Diagram

DFD Level 0 (context)

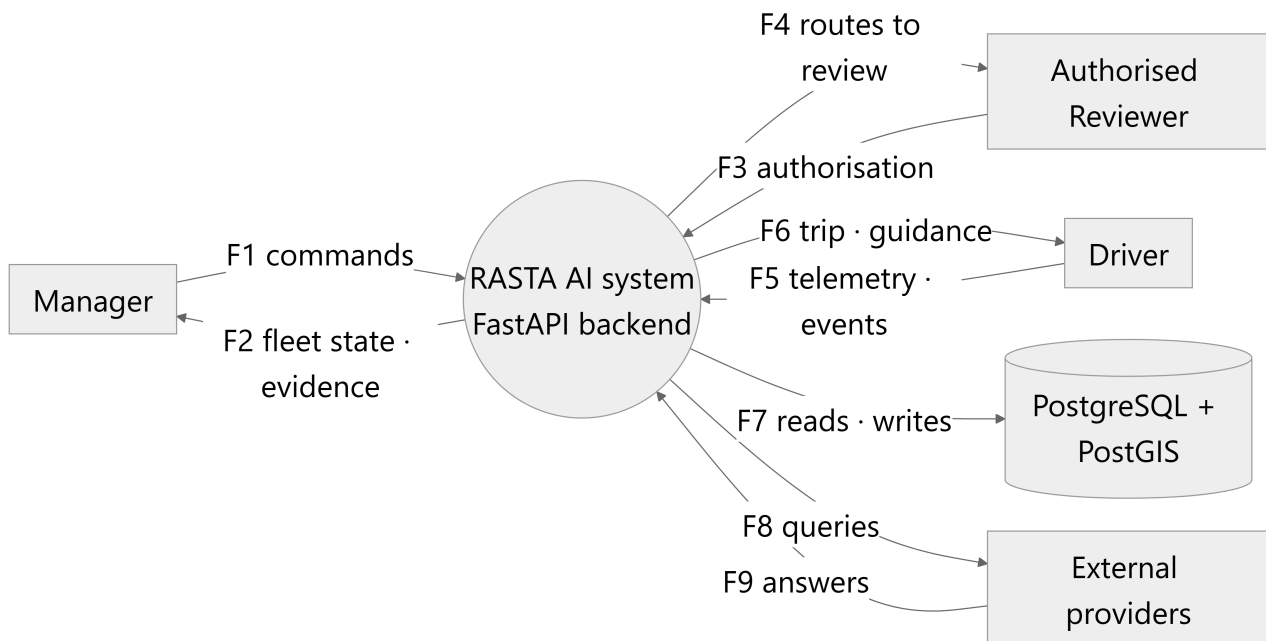


Figure 2 — Data flow diagram, Level 0 (context)

Flow	From → To	Content
F1	Manager → System	trips, assignments, route requests, route selection, dispatch, hold / cancel with disposition
F2	System → Manager	fleet snapshot (positions with age), observed tracks, route options with risk evidence, emergencies
F3	Reviewer → System	review authorisation for one selection, with rationale
F4	System → Reviewer	routes needing review with their evidence and reason codes
F5	Driver → System	accept, truck check (photo + plate), start, GPS batches, stop events, reroute request, instruction ACK, check-in, SOS
F6	System → Driver	current trip, offline package, navigation steps, route risk, notices, roadside places
F7	System ↔ Database	the backend is the only reader and writer
F8	System → Providers	route requests, weather / terrain / river / alert queries, geocoding
F9	Providers → System	route geometry and steps, observations, DEM samples, discharge, CAP alerts

DFD Level 1 (processes)

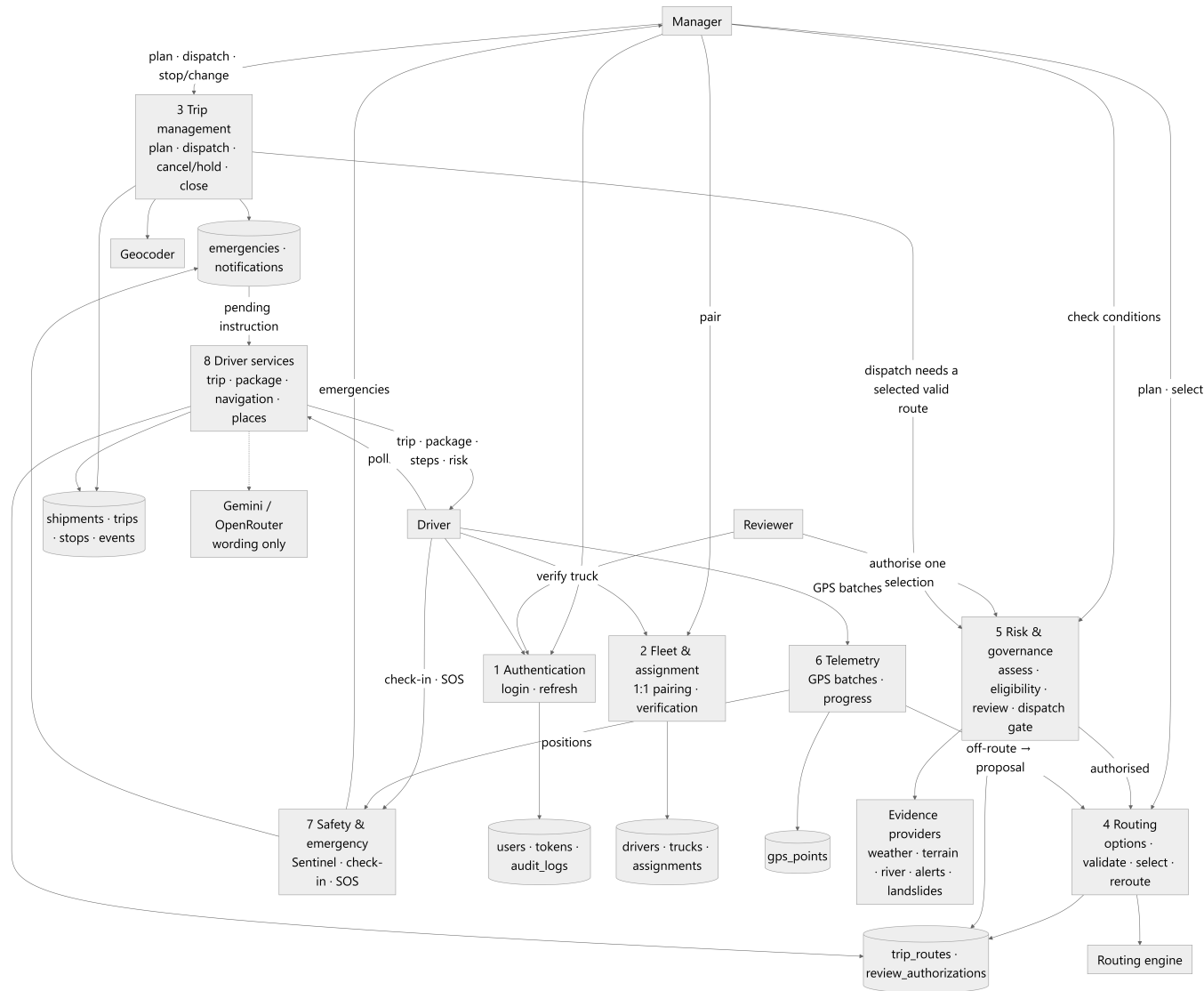
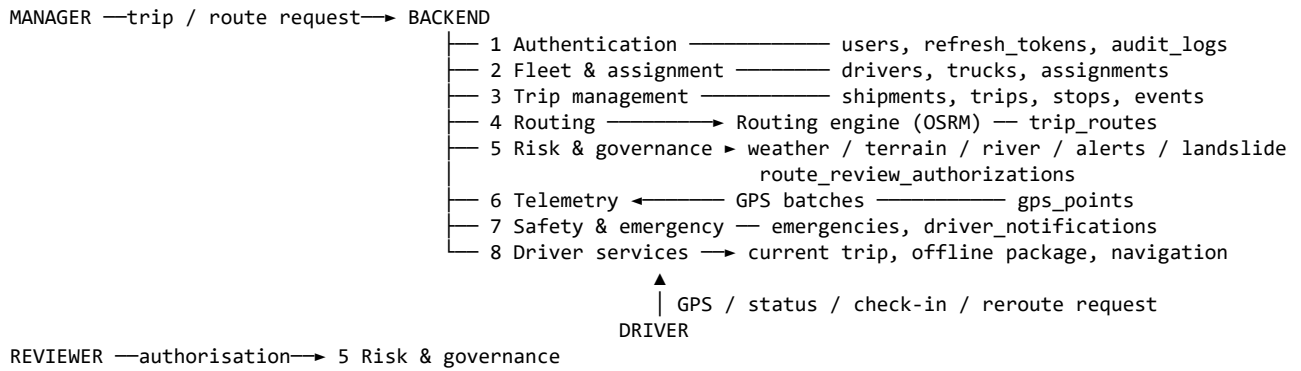


Figure 3 — Data flow diagram, Level 1 (processes, stores, external providers)

Text view of the same Level 1 flow:



8. Module Definition

Module	Responsibility	Input	Processing	Output	Dependencies	Status
Authentication	Sign in, refresh, identity	Identifier + password; refresh token	Argon2id verify; JWT issue; rotating refresh with reuse detection; rate limits	Access token, refresh (cookie or body), /me	users, refresh_tokens	IMPLEMENTED (services/auth.py, test_auth.py)
User / Role management	Roles and permissions	Role	Role → permission map; require_permission per handler	Allow / 403	Authentication	IMPLEMENTED (core/permissions.py, test_authorization.py)
Fleet records	Drivers, trucks, documents, maintenance	Manager forms	Validation, soft delete, document status	Records, profile drawers	Files	IMPLEMENTED (services/drivers.py, trucks.py; hosted UI 16 Sep)
Assignment	One driver ↔ one truck, verification	Driver id, truck id; driver photo + plate	Partial unique indexes; live-trip guard; verification / mismatch flag	Assignment record	Fleet records	IMPLEMENTED (services/assignments.py, migration 0006, 0011)
Trip management	Shipment + trip lifecycle	Client, cargo, confirmed stops, driver, truck	Capacity gate; region gate; atomic plan; dispatch gate; cancel with disposition; instructions + ACK	Trip, stops, events, notices	Routing, Governance	IMPLEMENTED (services/trips.py, test_post_pickup_resolution.py, test_shipment_trip_atomicity.py)
Route planning	Real road options	Confirmed endpoints (or a reported position)	Provider chain; endpoint + detour validation; distinct backups; supersede	trip_routes PROPOSED	OSRM	IMPLEMENTED (services/routes.py, domain/routing.py)
Route selection & reroute	Which road the trip follows	Route id, authorisation id	Eligibility refusal; spend authorisation in-transaction; reroute accept from position	SELECTED route; history	Governance	IMPLEMENTED (routes.apply_selection, services/reroute.py)
Navigation	Turn guidance	Selected route maneuvers, GPS	Next-turn selection; projection with hysteresis; spoken guidance	Next turn, distance, remaining	Driver map	IMPLEMENTED (driver-app/src/map/*, services/navigation.py)
GPS / Telemetry	Position ingestion	Fix batches	Idempotent insert; server time; progress projection (200 m)	gps_points; track; on/off-route	Trip	IMPLEMENTED (services/telemetry.py, domain/route_progress.py)
Fleet monitoring	Who is where, how fresh	Fleet snapshot	Contact freshness; stop progress	Map + table, inspector	Telemetry	IMPLEMENTED as 10 s polling (api/trips.py fleet router, useFleetPoll)
Weather	Observations along the route	Route points	Open-Meteo → MET Norway; freshness	Rain/wind components	—	IMPLEMENTED (services/weather/)
Terrain	Gradient profile	Route geometry	DEM sampling; class km; coverage floor	Terrain component, overlays	—	IMPLEMENTED (services/terrain.py, domain/terrain.py)
Risk analysis	Score and band	All evidence	11-factor points; bands; unavailable list; eligibility; recommendation	Assessment, decision	Weather, Terrain, Flood, Warnings, Landslide, Traffic	IMPLEMENTED (domain/route_risk.py, route_eligibility.py, route_recommendation.py)
Review authorisation	Human acceptance of incomplete evidence	Rationale	Bind evidence digest; expiry; single use; revoke	Authorisation row	Risk analysis	IMPLEMENTED (services/route_review.py, migration 0007)

Module	Responsibility	Input	Processing	Output	Dependencies	Status
Emergency (Fleet Sentinel)	Stationary / comms-lost / SOS	Positions, check-ins	Rules; deadline; escalation; frozen briefing; resolve	emergencies, manager panel	Telemetry	IMPLEMENTED; hosted end-to-end NEEDS TESTING; scheduler BLOCKED by config
Roadside services	Places near the corridor	Route, position	OSM snapshot query	Fuel, rest, repair, hospital	—	IMPLEMENTED (services/places/snapshot.py)
Offline data	Trip package on the phone	Selected/backup route, stops, guide	Fetch on start; 6 h freshness; stale label	Cached guidance	AsyncStorage	PARTIALLY IMPLEMENTED (no look-ahead prefetch on main; no offline tiles)
Driver assistant	Answers from state, in-language	Intent text	Keyword intents (offline); optional LLM wording via backend	Answer, guidance	i18n, Risk	IMPLEMENTED offline; online wording BLOCKED on hosted until API keys are set
Notifications	Notices, push relay	Events	Dedupe + cooldown; Expo push; local notifications	Driver notices	Expo	PARTIALLY IMPLEMENTED (device push BLOCKED: APK lacks google-services.json)
Route watch worker	Conditions ahead of a moving truck	Positions, evidence	Corridor window by speed; material-change detection	Push events	Risk analysis	IMPLEMENTED (services/route_watch.py), enabled on Render; NEEDS TESTING with device push
Files & documents	Private photos and documents	Uploads	Type/size validation; owner-or-manager reads; masked numbers	stored_files	Auth	IMPLEMENTED (api/files.py, documents.py, test_files_documents.py); no OCR by design
Database layer	Persistence, geometry, audit	ORM models	SQLAlchemy 2.0 async; PostGIS geography; Alembic; audit rows	Tables	Supabase	IMPLEMENTED (12 migrations, test_migrations.py, test_schema_drift.py)
External provider integration	Health and fallbacks	Provider calls	Timeouts; chains; health rows	/api/system/providers	—	IMPLEMENTED (services/provider_health.py)
Demo simulation	Labelled scenario injection for demos	Scenario name	Adds codes/points to the real assessment	Labelled risk	Risk analysis	IMPLEMENTED, demo-only (DEMO_SIMULATION_ENABLED)
Payments / PoD / payroll	Financial closure	—	—	—	—	NOT IMPLEMENTED (design only, docs/DATA_MODEL.md)
WebSocket live channel	Server push to consoles	—	—	—	—	NOT IMPLEMENTED (design in docs/ARCHITECTURE.md; polling shipped)
Landslide ML model	Probability of slide	—	—	—	—	BLOCKED_BY_DATA (docs/AI_MODELS.md); research scripts only
SMS fallback	Manager alert without data	—	—	—	—	NOT CONFIGURED (SMS_PROVIDER unset)

9. Technology Stack

Versions are read from `manager-web/package.json`, `driver-app/package.json`, `backend/requirements.txt` and the installed environment; nothing below is guessed.

Frontend — Manager

- **Framework:** React 19.2 (react, react-dom), React Router 7.18
- **Language:** TypeScript 6.0
- **UI libraries:** Tailwind CSS 4.3 (@tailwindcss/vite), lucide-react icons
- **Map library:** MapLibre GL 6.6 with OpenStreetMap raster tiles
- **Testing:** Vitest 3.2, Testing Library (react 16.3, user-event 14.6, jest-dom 7), jsdom 30 — **225 tests passing** (16 Sep 2026)
- **Build system:** Vite 8.2, tsc -b, oxlint 1.79
- **Hosting:** Render static site, vite build --mode remote-demo

Driver Application

- **Framework:** Expo SDK 57 (expo ~57.0.18), React Native 0.86.3, React 19.2.3, react-native-web 0.21 for the web build
- **Language:** TypeScript 6.0
- **Navigation:** in-app tab navigation (src/navigation.ts); map via Leaflet 1.9 inside react-native-webview 13.16 (native) / Leaflet directly (web); guidance from OSRM step data
- **Location / GPS:** expo-location 57 (single watcher, foreground)
- **Storage / cache:** @react-native-async-storage/async-storage 2.2 (GPS queue, trip package, language), expo-secure-store (tokens)
- **Other:** expo-notifications (local + push token), expo-speech (spoken guidance), expo-image-picker / expo-document-picker (truck photo, documents), expo-intent-launcher (hands speech input to the system recogniser); emergency calls use tel: links so the dialler places the call
- **Testing:** Vitest 4.1, jsdom — **624 tests passing** (16 Sep 2026)
- **Build:** EAS Build, profile remote-demo → Android APK 1.0.18 (versionCode 18); check-release-config.mjs guard

Backend

- **Framework:** FastAPI 0.115.6 on Uvicorn 0.34 (Starlette 0.41)
- **Language:** Python 3.11.9 (pinned >=3.11, <3.13)
- **ORM / data access:** SQLAlchemy 2.0.36 (async) over psycopg 3.2.3, GeoAlchemy2 0.16, Alembic 1.14 (12 migrations)
- **Validation:** Pydantic 2.10 + pydantic-settings 2.7
- **Authentication:** argon2-cffi 23.1 (Argon2id), PyJWT 2.10 (HS256; crypto extra ready for RS256/JWKS)
- **HTTP client:** httpx 0.28 (providers, tests)
- **API architecture:** REST, /api/*, 14 routers, Pydantic schemas, uniform error envelope, permission dependency per handler, in-process rate limiter
- **Testing:** pytest 8.3, pytest-asyncio 0.25, pytest-cov — **1172 passed, 5 skipped** on an isolated PostgreSQL cluster (16 Sep 2026)
- **Container / hosting:** Dockerfile on Render (free, Singapore region), buildFilter on backend/**

Database

- **Database:** Supabase PostgreSQL 17.6 (ap-south-1); optional local PostgreSQL 18 for offline development; isolated test cluster on port 55432
- **Geospatial support:** PostGIS 3.3 — geography(Point/LineString, 4326), ST_DWithin, route length in metres
- **Authentication integration:** Application-level JWT (Supabase Auth verifier scaffolded, not enabled)
- **Security:** RLS enabled on every table (no policies), backend is the only writer, SSL required, pool sized for the session pooler

Mapping / Routing

- **Map rendering:** MapLibre GL (manager), Leaflet (driver), OSM tiles
- **Routing provider:** OSRM (ROUTING_PRIMARY_URL optional, public OSRM fallback), 8 s timeout, up to 3 options with steps
- **Geospatial processing:** PostGIS on the server; haversine, polyline projection, hysteresis and gradient in pure TypeScript/Python domain code (parity-tested)

Weather / Risk Sources

- **Providers:** Open-Meteo (weather, elevation, flood/GloFAS), MET Norway (weather fallback), OpenTopoData (elevation fallback), NDMA SACHET CAP (official alerts), NASA Global Landslide Catalog (bundled snapshot), Nominatim (geocoding), own fleet probes (traffic)
- **Usage:** sampled along the route at assessment time; cached with TTLs (warnings 10 min)
- **Fallback:** weather and terrain have a second provider; the rest degrade to UNKNOWN / NOT_AVAILABLE

AI providers

- Gemini Developer API (primary), OpenRouter free models (fallback): wording only, 10 RPM cap, prompt with facts supplied by the backend; Ollama client present, no model installed

DevOps / Tooling

- **Git / GitHub:** main branch, 106 commits; draft PR #1 (connectivity/prefetch) unmerged
- **Testing:** three suites above; hosted rehearsal scripts (.runtime/rehearsal/*.mjs, CDP-driven Chromium; phone via ADB + uiautomator)
- **Build:** Vite, tsc, EAS Build, Docker
- **Environment management:** .env per app, pydantic-settings, render.yaml blueprint, docker-compose.yml for local Postgres
- **CI:** NOT VERIFIED (no CI workflow file inspected in this document's evidence; tests are run locally and recorded in HANDOFF.md)
- **Deployment:** Render (backend Docker web service, two static sites), auto-deploy on push to main

10. Technology Justification

Technology	Why this technology	Why it fits this project	Advantages	Limitations	Alternatives considered
FastAPI + Pydantic	Typed request/response contracts, async I/O, automatic OpenAPI	Many small provider calls in parallel; contracts shared with two clients (docs/API_CONTRACTS.md)	Fast to write and test; validation at the boundary is the safety layer for coordinates	Single-process; no built-in job queue	Django REST (heavier), Node/Express (team's Python evidence code)
PostgreSQL + PostGIS on Supabase	Managed Postgres with geography types	Distances in metres for proximity rules; one system of record	Free tier, PITR, ap-south-1 latency	Session pooler limits connections (15); no RLS policies used	MongoDB (no geography semantics), SQLite (no PostGIS)
SQLAlchemy 2.0 async + Alembic	Explicit transactions and row locks; versioned schema	Trip row locks stop double dispatch and stale selection (test_stale_selection_interleaving.py)	Mature; migrations tested for drift	Verbose	Prisma/Drizzle (TypeScript backend not chosen)
React 19 + Vite + Tailwind	Fast build, component tests	Console with many state-driven controls	Vitest + Testing Library cover 225 cases	Bundle > 500 kB (MapLibre)	Next.js (SSR unnecessary), Angular
MapLibre GL	Open, no key, vector/raster	Planned vs observed layers, hazard overlays	No licensing surprise	Raster OSM tiles depend on network	Google Maps JS (key, cost), Leaflet (used on driver)
Expo + React Native	One TypeScript codebase for Android and web	Web build doubles as a judge fallback and as the GPS-simulation harness	EAS builds without a local Android toolchain	Native map is a WebView; no background location	Flutter, native Kotlin
Leaflet in WebView (driver)	Lightweight, no key	Works on the web build identically	Simple overlays	No offline tiles; WebView bridge	Mapbox/Google SDKs (keys, size)
OSRM	Open routing with steps and alternatives	Real roads for NER; keyless fallback exists	Free; interface allows a self-hosted primary	Public demo server has no SLA; car profile, not truck	Valhalla (truck profile; self-host later), Google Directions (cost)
Open-Meteo / MET Norway / OpenTopoData / GloFAS	Free, keyless, documented	Evidence must be affordable to sample at 10 points per route	Two weather providers; two DEM providers	Rate limits; forecasts not ground truth	IMD APIs (access), paid weather
NDMA SACHET CAP	The official Indian alert feed	Alerts are placed by district onto the corridor	Authoritative	District granularity only	None equivalent
Argon2id + JWT with rotating refresh	Current password-hashing recommendation; stateless access tokens	Short access life limits token theft; reuse detection catches replay	Standard, testable	Secret rotation is manual	Session cookies only; Supabase Auth (scaffolded for later)
Polling (10 s) instead of WebSocket	Simplicity and resilience on Render free tier	Bounded, cache-friendly, survives cold starts	Simple failure semantics (last good state)	10 s latency; not "real-time"	WebSocket (designed, not built)
Deterministic rules over ML	No calibrated dataset exists	Judges can read every threshold; UNKNOWN is honest	Auditable; testable (test_reason_code_coverage.py)	No learned improvement	Landslide model (blocked by data)
Render + Docker	Deploys a Dockerfile from GitHub with no CLI	Hosted certification is the actual submission	Free; blueprint in repo	Cold starts ~30 s; restarts on deploy	Fly.io, Railway, Cloud Run (same image works)

11. Scalability Analysis

CURRENT HACKATHON SCALE	4 drivers, 4 trucks, one corridor at a time, Render free tier, Supabase free tier
↓	
MULTIPLE VEHICLES	tens of trucks: GPS at 10 s moving → ~6 fixes/min/truck, batched (6 per POST); idempotent insert; fleet snapshot is one query per 10 s per console
↓	
MULTIPLE FLEETS	tenancy is not modelled today (one organisation); would need an org id on users/drivers/trucks/trips and permission scoping – NOT IMPLEMENTED
↓	
DISTRICT / STATE	evidence fan-out becomes the cost: 10 weather points per route per check; route watch worker per active trip every 60 s – cache by corridor cell
↓	
MULTI-REGION DEPLOYMENT	stateless API behind a load balancer; Postgres read replicas for the fleet snapshot; per-region routing engine

Concern	Today	Next step
Backend scaling	One Uvicorn process on Render free; stateless except the in-process rate limiter and warnings cache	Horizontal replicas; move rate limiting and caches to Redis; separate worker process for route watch and Sentinel
Database scaling	Pool 3 + 2 (Render) against Supabase's 15-client session pooler; indexes on gps_points(trip_id, device_fix_id), assignments	Transaction pooler (PgBouncer) for API; partition gps_points by month; read replica for snapshots
GPS event volume	100 trucks × 6 fixes/min = 600 rows/min, batched into ~100 POSTs/min	COPY-style bulk insert; retention job (design in docs/SECURITY.md §10)
Route-provider limits	Public OSRM demo server, no SLA, 8 s timeout	Self-hosted OSRM/Valhalla with NER extract and truck profile
Weather-provider limits	Open-Meteo free tier; 10 samples per route per check; checks only on request, never polled	Corridor-cell cache with TTL; batch endpoints; MET Norway spillover
Caching	Warnings feed 10 min TTL; landslide/places snapshots bundled; client keeps last good reading	Shared cache (Redis) for assessments keyed by route id + evidence version
Rate limiting	Per-IP and per-identifier login/refresh limits in process	Distributed limiter; per-role API quotas
Observability	/health, /ready, /api/system/providers with provider health rows; audit log	Structured logs to a sink, request ids in error envelope (already present), metrics (latency per provider)
Background processing	Route watch and warnings poller as in-process async tasks; Sentinel sweep on demand	Dedicated worker with a queue; Sentinel scheduler enabled
Offline synchronisation	Durable bounded GPS queue; idempotent server	Conflict-free stop events with client ids (stops already sequenced server-side)

12. Security

Area	What is in place	Evidence
Authentication	Argon2id hashes; 15-minute access JWT; 30-day rotating refresh with family revocation on reuse; web gets an HttpOnly refresh cookie (SameSite=None across Render subdomains), native gets it in the body	services/auth.py, api/auth.py, docs/SECURITY.md §1
Authorization	4 roles → 32 permissions enforced by a dependency on every mutating handler; reviewer cannot select routes; driver cannot open the manager console ("Manager account required"); manager cannot see driver passwords	core/permissions.py, test_authorization.py, hosted check 16 Sep
RLS	Enabled on all 20 tables with no policies; the anon key reads nothing; backend is the only writer	docs/DATA_MODEL.md §1, test_rls_boundary.py
Secret management	SECRET_KEY generated by Render; DATABASE_URL, GEMINI_API_KEY, OPENROUTER_API_KEY are dashboard secrets (sync: false); .runtime/ and .env* are git-ignored; the Supabase publishable (anon) key in eas.json is public by design and grants no data access under RLS	render.yaml, .gitignore
API keys	No routing/weather keys needed; MapTiler key (if used) only through environment; Google Places optional	core/config.py
Input validation	Coordinate bounds (PostGIS silently wraps inverted latitude), service-region bbox, cargo ≤ capacity, reason ≥ 10 chars, rationale ≥ 20 chars, disposition enum, reroute origin plausibility	schemas/common.py, schemas/domain.py, test_geospatial.py
Rate limiting	Login 20/IP and 10/identifier per 60 s; refresh 60/IP; 429 with retry semantics	core/rate_limit.py, test_rate_limit.py
Sensitive data	Driver document numbers masked; private files served owner-or-manager only with audit DOCUMENT_ACCESS; no Aadhaar OCR; no government verification claims	api/files.py, api/documents.py, test_files_documents.py
GPS privacy	Location is uploaded only while a trip is active and the app says so; fleet:location_read is a separate permission the reviewer lacks; retention policy documented	docs/SECURITY.md §3, §10
Manager / driver permissions	Manager: fleet, trips, routes (plan/select), dispatch; Driver: own trip, own location; Reviewer: read + authorise only	core/permissions.py
Transport	HTTPS on Render; CORS restricted to the two static origins; SSL required to the database	render.yaml, main.py

Known gaps (documented, not hidden): no CI-enforced secret scanning; Supabase Auth not enabled (local JWT only); no per-organisation tenancy; retention job not scheduled (docs/SECURITY.md §12).

13. Failure & Degradation Strategy

Failure	System response	Driver impact	Manager impact	Recovery
Internet failure (phone)	Poll fails → isStale; last trip state kept; GPS queued durably (bounded)	Guidance from cached package (≤ 6 h); Safety tab offline; "Last sync Ns ago" grows; no map tiles	Contact freshness ages to stale then NO CONTACT; Sentinel comms-lost rule after 60 min	Next successful poll clears stale; queue flushes in order; server de-duplicates
GPS failure	Watcher error → isTracking=false; last-known shown with age; stationary filter	"GPS off" chip; no invented position	Last position with age; LIVE never claimed	Automatic when fixes resume
Routing provider failure	Chain fallback; 8 s timeout; 503 ROUTING_UNAVAILABLE	Existing selected route unaffected; reroute request fails with a message	"Plan route" shows a retryable error; nothing stored	Retry; configure ROUTING_PRIMARY_URL
Weather provider failure	MET Norway fallback; then factor NOT_AVAILABLE	Card shows "N of M factors"	Coverage line reads "Weather UNKNOWN"; band from remaining evidence	Automatic on next check
Backend unavailable (cold start / restart)	15 s client timeout (90 s for evidence reads); "Cannot reach the backend" / "took too long" beside the control; unreadable 2xx body treated as error	App keeps last state; queue grows	Last good snapshot with banner; Try again repeats the failed action	Render restarts in ~30 s; buildFilter stops restarts on unrelated pushes
Battery low	Stationary interval 60 s; background poll 3× slower; one watcher	Longer gaps between uploads	Slightly older positions	—
Stale GPS	Server judges age on server time; quantised age on phone	"Last known · N min"	Row turns stale → NO CONTACT	Fresh fix
Invalid route (provider answer does not match the question)	endpoint_mismatch / detour rule → 422 ROUTE_VALIDATION_FAILED; nothing stored	None	Clear refusal message	Re-plan or fix the stops
Database error	5xx with request id; transaction rolled back; authorisation never spent on a failed selection	Poll fails → stale	Error state with retry	Supabase recovery; /ready reports database checks
Authentication expiry	One silent refresh on 401; failure → login screen; refresh reuse revokes the family	Re-login	Re-login	Token rotation
Evidence provider rate-limited (429)	Provider health row RATE_LIMITED; factor unavailable	Same as provider failure	Diagnostics shows the provider state	TTL expiry
Push relay unavailable	Local notifications + polling	Notices arrive on next poll (≤ 10 s)	—	Add google-services.json and rebuild

Controlled degradation is the design rule: the system narrows what it claims rather than pretending. UNKNOWN, NOT_CHECKED, NOT_MEASURED, NOT_CONFIGURED are rendered states, not errors.

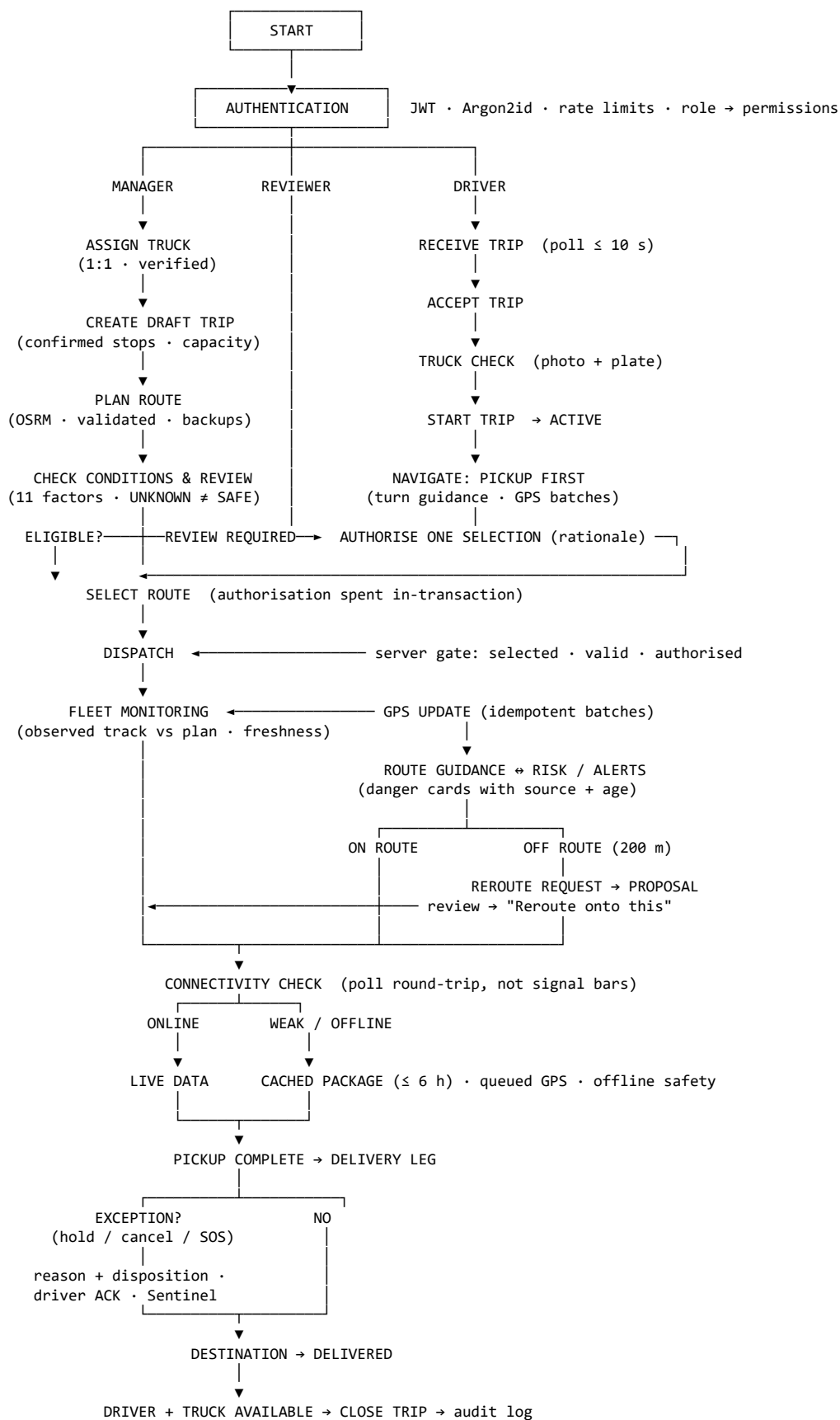
14. Implementation Status Matrix

Capability	Status	Evidence	Remaining work
Manager login, RBAC, driver-on-manager refusal	IMPLEMENTED	Hosted 16 Sep; test_authorization.py	—
Driver/truck records, documents, profile drawer	IMPLEMENTED	Hosted UI audits §14, §17	—
Assign truck (1:1, live-trip guard, verification)	IMPLEMENTED	assign_truck_check.mjs 13/13; §17	—
Trip planner with confirmed locations and region gate	IMPLEMENTED	planValidation.test.ts, server validator, §12	—
Atomic shipment + trip creation	IMPLEMENTED	test_shipment_trip_atomicity.py	—
Route planning with validation and real backups	IMPLEMENTED	test_route_api.py, hosted 2.9 s	Self-hosted truck-profile router
11-factor risk assessment	IMPLEMENTED	test_route_risk.py family; hosted 5–9 s	Current landslide inventory
Eligibility + reviewer authorisation + dispatch gate	IMPLEMENTED	test_dispatch_route_gate.py (8), test_route_review_authorization.py; hosted §16–§17	—
State-driven route control (Use / Open review / Blocked)	IMPLEMENTED	TripRouteReview.test.tsx (9), hosted §16	—
Dispatch → phone receipt	IMPLEMENTED	Physical phone ≤ 9 s, §17	—
Accept, truck verification, start	IMPLEMENTED	Physical phone §7, §17	Manager has no "accepted" signal (P3)
Pickup-first navigation, turn guidance	IMPLEMENTED	Physical phone §17; maneuvers.test.ts	Offline tiles
GPS batching, idempotency, stationary filter	IMPLEMENTED	tracker.test.ts, test_telemetry.py	Cached fix shown as "GPS live" when device location is off (P3)
Observed track vs planned corridor	IMPLEMENTED	Hosted §11, §17	—
Off-route → reroute proposal → review → accept → phone	IMPLEMENTED	Hosted simulated GPS + phone §17	Inspector must reload to see a new proposal (P3); far-origin guard added 16 Sep
Post-pickup cancel/hold with disposition and driver ACK	IMPLEMENTED	test_post_pickup_resolution.py (10), hosted smoke §13	—
Delivery → driver/truck AVAILABLE → close	IMPLEMENTED	Hosted §17	—
Fleet Sentinel (stationary / comms-lost / check-in / SOS)	IMPLEMENTED, NEEDS TESTING (hosted e2e)	test_sentinel*.py, test_emergency_api.py	Enable scheduler; hosted escalation run
Roadside places	IMPLEMENTED	test_places_snapshot.py; phone "Find a place to stop"	Live POI provider
Offline trip package (6 h)	IMPLEMENTED	test_offline_package.py, packageStore.test.ts	—
9 km look-ahead prefetch	NOT IMPLEMENTED (on main)	Draft PR #1 unmerged	Review and merge PR #1
Offline map tiles	NOT IMPLEMENTED	—	Tile cache in WebView
Connectivity truth from poll success	IMPLEMENTED	TripProvider.tsx; navigator.onLine not used	—
Multilingual UI	PARTIALLY IMPLEMENTED	appLanguage.ts statuses	Verify DRAFT languages with native speakers
Driver assistant (offline intents)	IMPLEMENTED	assistant.test.ts (no network import)	—
Online AI wording (Gemini/OpenRouter)	IMPLEMENTED, BLOCKED on hosted	test_gemini_ai.py; Diagnostics shows keys unset	Set API keys on Render

Capability	Status	Evidence	Remaining work
Push notifications to device	BLOCKED	Server relay implemented (migration 0012); APK lacks <code>google-services.json</code>	Add Firebase config, rebuild APK
SMS fallback	NOT CONFIGURED	<code>SMS_PROVIDER</code> unset; provider health row	Choose a gateway
WebSocket live updates	NOT IMPLEMENTED	Design only; polling shipped	—
Payments, proof of delivery, payroll	NOT IMPLEMENTED	<code>docs/DATA_MODEL.md</code> design-only	Future phase
Landslide ML model	BLOCKED_BY_DATA	<code>docs/AI_MODELS.md</code> §0a; research scripts only	Obtain a current inventory
Trips older than the newest 50 in the console	PARTIALLY IMPLEMENTED	\$17 open item	Pagination / search on Trips
APK carries latest driver-web fixes	PARTIALLY IMPLEMENTED	APK 1.0.18 predates two driver changes of 16 Sep	Rebuild only if a phone blocker appears
CI pipeline	NOT VERIFIED	No workflow file inspected	Add CI running the three suites

Test baseline (measured 16 Sep 2026, not inherited): backend 1172 passed / 5 skipped; manager 225 passed; driver 624 passed; typecheck and production builds clean. Hosted certification: `bash .runtime/judge.sh check` → READY.

15. Final System Flowchart



Nodes present in the original template but replaced here: "PLAN ROUTE before ASSIGN DRIVER" (in RASTA AI the pairing exists before the trip and is auto-filled into the planner), the implicit "route is safe" step (replaced by the evidence check and the reviewer branch), and any real-time push arrow (the system polls).

16. Completion Report

REQUIRED HEADINGS: 9/9

1 Problem Statement Understanding	\$1
2 Proposed Solution	\$2
3 Unique Key Features	\$3
4 Target Users	\$4
5 System Architecture	\$5
6 Workflow / User Flow Diagram	\$6
7 Data Flow Diagram	\$7
8 Module Definition	\$8
9 Technology Stack	\$9

TECHNICAL CLAIMS VERIFIED:

Every status label in \$2, \$3, \$8 and \$14 is tied to a file, a test, a migration, a configuration key, or a dated hosted-certification log (HANDOFF.md \$7-\$17).
Versions in \$9 are read from package.json / requirements.txt / pip list.
Test counts are the 16 Sep 2026 runs (backend 1172/5 skipped, manager 225, driver 624).

UNVERIFIED CLAIMS:

CI pipeline (no workflow inspected) – marked NOT VERIFIED.
Hosted end-to-end run of a Sentinel escalation – marked NEEDS TESTING.
Route watch worker with device push – marked NEEDS TESTING (push is BLOCKED).
No accuracy figure of any kind is claimed.

IMPLEMENTED FEATURES:

Auth/RBAC · fleet records · 1:1 assignment with physical verification · atomic trip planning with confirmed locations and region gate · validated route planning with real backups · 11-factor evidence risk (UNKNOWN ≠ SAFE) · eligibility + reviewer authorisation + server dispatch gate · state-driven route control · dispatch → phone · accept / truck check / start · pickup-first turn guidance · batched idempotent GPS with durable queue · observed track vs plan · off-route → governed reroute → phone follows · post-pickup cancel/hold with disposition and driver ACK · delivery and close · roadside places · offline trip package · offline assistant, safety guide, phrasebook · connectivity truth from poll success · Fleet Sentinel rules and emergencies API · provider health diagnostics · demo simulation.

PARTIALLY IMPLEMENTED:

Offline strategy (package yes; look-ahead prefetch and tiles no) · multilingual UI (English verified, others DRAFT/FALLBACK) · notifications (server relay yes, device push blocked) · trips list limited to newest 50 · APK predates two driver-web fixes.

NOT IMPLEMENTED:

WebSocket live channel · payments / proof of delivery / payroll · offline map tiles · 9 km prefetch on main · SMS fallback (not configured) · landslide ML model (blocked by data) · multi-organisation tenancy.

DIAGRAMS CREATED:

High-level architecture (Mermaid, \$5.1) · DFD Level 0 (Mermaid, \$7) · DFD Level 1 (Mermaid + text, \$7) · Manager workflow (\$6.1) · Driver workflow (\$6.2) · Emergency flow (\$6.3) · Offline strategy ladder (\$2.6) · Scalability ladder (\$11) · Final system flowchart (\$15) · four persona flows (\$4).

FILES CREATED:

docs/submission/day1/RASTA_AI_SIH2026_DAY1_TASK1_SYSTEM_DESIGN.md
docs/submission/day1/RASTA_AI_SIH2026_DAY1_TASK1_SYSTEM_DESIGN.docx
docs/submission/day1/RASTA_AI_SIH2026_DAY1_TASK1_SYSTEM_DESIGN.pdf
docs/submission/day1/diagrams/*.png (rendered Mermaid diagrams for the DOCX/PDF)

MAIN TECHNICAL RISKS:

1. Public OSRM demo server has no SLA (mitigation: configurable primary; self-host).
2. Render free tier cold starts and restarts (mitigation: buildFilter added; clients retain last good state; 90 s budget for evidence reads).
3. Landslide inventory ends in 2017 → every corridor is REQUIRES_REVIEW on hosted (mitigation: the reviewer step is the demo's governance feature, not a bug; a current inventory would raise ELIGIBLE rates).
4. Device push blocked without Firebase config (mitigation: polling ≤ 10 s; local notifications).
5. One session per driver account: a second sign-in signs the first out (documented).
6. Supabase session pooler's 15-client limit caps API concurrency (mitigation: small pool; transaction pooler next).

MENTOR QUESTIONS WE SHOULD PREPARE FOR:

- "Where is the AI?" → evidence-based deterministic decision support; LLMs word, never decide; the ML model is blocked by data and we say so (docs/AI_INVENTORY.md).
- "What happens with no network for an hour?" → \$2.6 and \$13, with the 6 h package and the Sentinel comms-lost rule; and what is NOT there (tiles, prefetch).
- "Why does every route need review?" → no landslide inventory configured → UNKNOWN ≠ SAFE; the reviewer accepts incomplete evidence for one dispatch, with a rationale, audited.
- "Is tracking real-time?" → near-live polling at 10 s; measured latencies in \$3.15.
- "How do you stop a wrong route?" → endpoint/detour validation, region gate, reroute origin plausibility (the 2,908 km and 2,451 km incidents and their fixes).
- "How does this scale to a state?" → \$11 ladder; what is stateless today; tenancy gap.
- "What is unfinished?" → \$14 verbatim.